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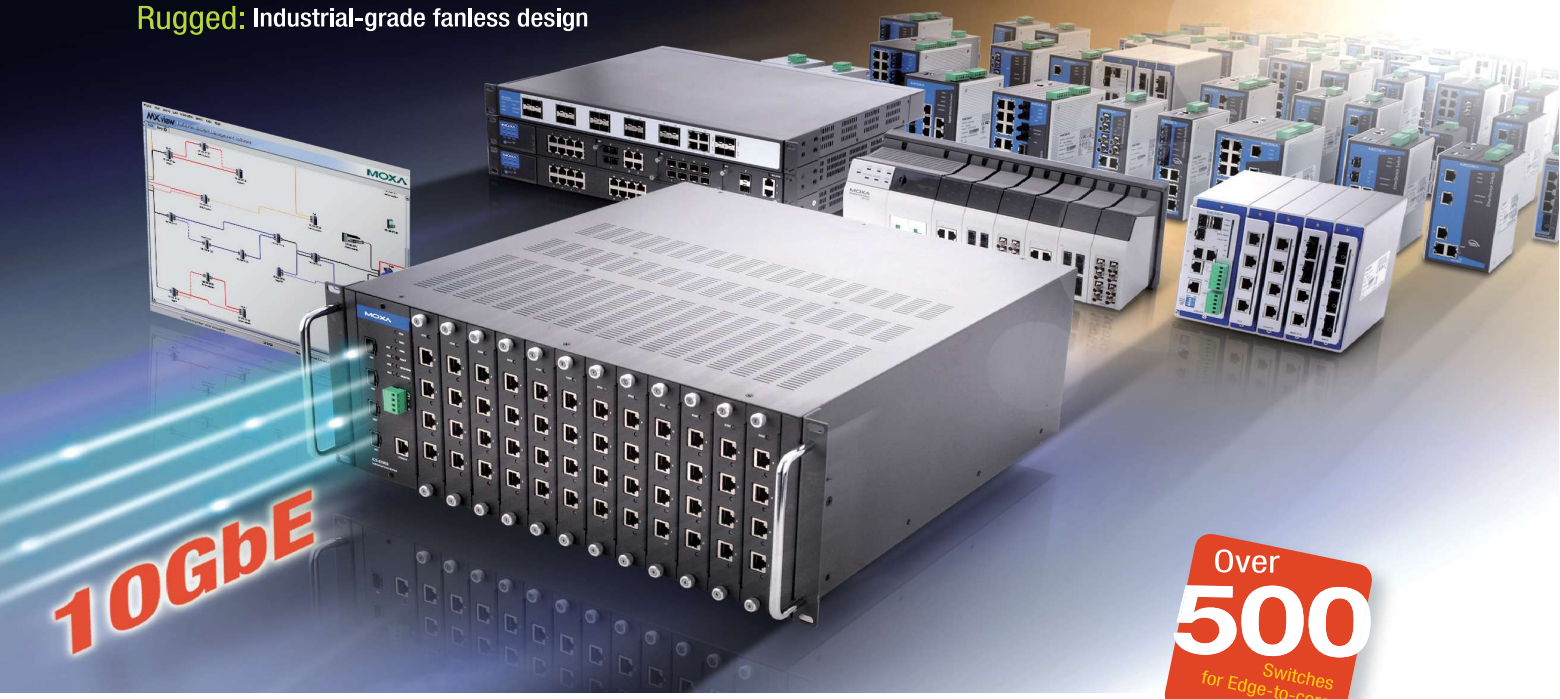

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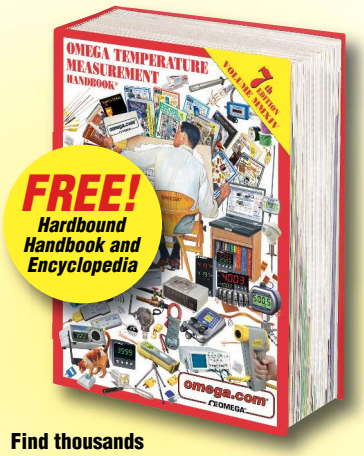

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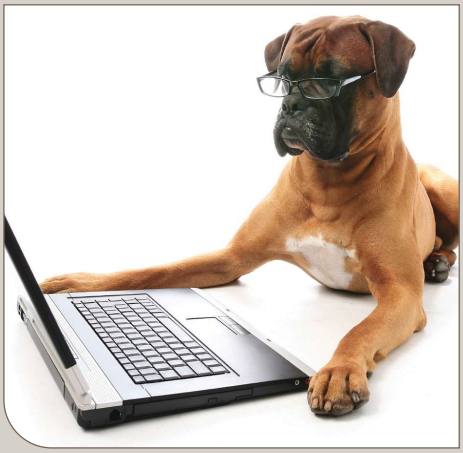
WEB EXCLUSIVE

OMAC Packaging Workgroup

What do "dog years" and industrial standards development have in common? It's not because standards are man's best friend, which they are. It's about the need for resolve and persistence to see standards implemented, and that's the OMAC story. Read John Kowal's feature at **www.isa.org/intech/201110web**.



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Smart Grid avoiding the real challenges?

By Bill Lydon, *InTech*, Chief Editor

In my opinion, there is too much hype about the Smart Grid without looking at the core problems from an engineering and business point of view. There is a constant flow of information, presentations, and discussions about the Smart Grid, but I wonder if this simply makes people feel good and avoids facing the real issues. Even politicians are rallying around Smart Grid buzzwords; I call this kind of discussion "cocktail party technology talk." There is no doubt electrical energy can be managed better, and the Smart Grid concepts will provide more data and can be used to control the flow of energy from generation to users. This does not address the heart of the issue.

The inescapable fact is the demand for electricity is going to be larger than generation capacity in place today, requiring some real planning to avoid problems. At the highest level, there are only two key factors concerning the electrical energy problem namely, demand and supply. The amount of growing demand for power relative to the supply available is a serious problem. The economic slowdown has moderated demand, but this will change as growth returns. Ultimately, there are only two ways to meet future growing energy requirements—increase generation capacity and lower consumption.

Creating more electrical power takes time and investment at a pace that is beyond today's commitments to build more electrical power generation. This encompasses traditional and alternative power generation, including wind and solar. Adding more capacity, in most cases, requires the addition of more power lines to deliver electricity to users. Major investments need to be made installing electric transmission lines and distribution transformers in emerging countries and replacing aging transmission systems in the U.S. and Europe where about 75% of the networks are more than 30 years old.

The Smart Grid can be a contributor helping better manage the flow of electricity and possibly demand control. Generally, demand control described by Smart Grid experts is deferral of demand to avoid power peaks. Deferring demand is not saving energy but putting off the time some energy will be consumed. This does lower peak power generation requirements but does not solve the growing need for more power. I saw a presentation on a novel research project to defer demand in a vehicle-to-grid demonstration project that utilizes stored power in an electric car's battery to feed the electric grid while it is connected to a charging station to essentially help lower peak power generation requirements. In the example, it was noted that when people come home from work, they plug in the car for charging (initially the car's battery is used as a power source) and turn on the home air-conditioning. Interesting approach, but the batteries still need to be charged before the car needs to be used.

The most effective near-term solution is energy conservation to use less electricity. Obviously, conservation lowers the requirement to have more generation capacity. Automation professionals can have a direct impact on energy conservation looking at three basic areas for improvement. First, save energy by running plants and processes more efficiently using better controls and optimization. Second, replace older devices with more efficient ones; for example, replace old motors with high-efficiency models. Third, redesign process and production lines to consume less energy.

Why wait for the Smart Grid; let's be smart today and find ways to save energy. If we all believe the demand and cost of energy is going to increase over time, we should be willing to apply energy conservation methods that will lower our cost and risk. Saving energy improves profits and is protection against higher energy prices in the future.

Please share your thoughts at blydon@isa.org.

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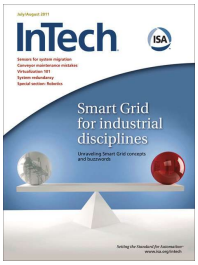
Get to know people

When I first started as an engineer, I had an aversion to networking. In fact, I was downright ALLERGIC to it.



So, what changed? For one thing, I realized rather quickly that the engineering community here thinks like a small town—everyone knows everyone. If I hoped to gain (and retain) employment, I'd better get to know some people. The other thing that happened was that I became more confident as I advanced in my profession. Although I still dislike crowds, I can and will carve out small groups to speak with at an event. (One thing I've been doing to improve that skill is attending a women's networking event every month.)

Another good tool is social networks. You can make a lot of long distance professional acquaintances on LinkedIn and Facebook. But ... social networking via computer is not and cannot ever completely replace the need for meeting people face to face.

Karen D. Morton, P.E.

Letter in reference to July/August InTech *"The Final Say"*

The complete package

In 1968, GM wrote a specification for a programmable control system, which could replace relay logic. The timing for this spec was right, as a group of designers led by Dick Morley was already working on a design for what became the Modicon 084. GM purchased this Programmable Controller (later known as Programmable Logic Controller), and the PLC is now an immeasurably important technology throughout the world of manufacturing.

PLCs and Distributed Control Systems (DCS) routinely interact with plant operators through computer-based, graphic, Human Machine Interfaces (HMI). The development of the HMI has evolved over a number of operating systems including unique proprietary systems, Unix, BSD, DOS, Windows, and Linux. The most widely used HMIs and arguably the most advanced run on Windows.

One difficulty presented by the Windows operating platform is the rate at which the OS changes. HMI developers exhibit so much time lag while trying to release stable updates that they can easily be an entire release behind.

As a systems integrator, I can say that we are currently (August 2011) supporting the new installation of systems on Windows XP. Last week, we had to scramble to find a copy of Windows 7 with no service packs. Service Pack 1 has already been released, so this will become more difficult over time.

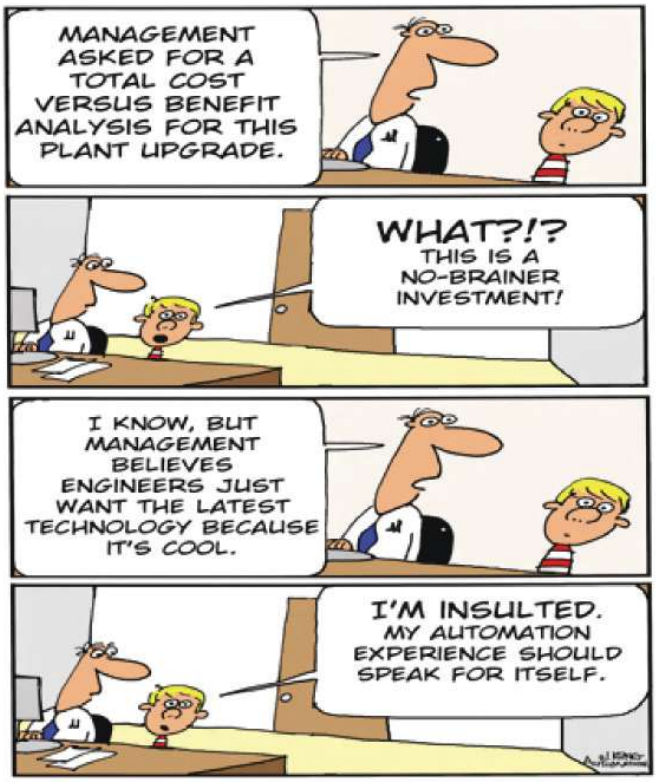
Now, I believe the time is right for a large corporation to write a specification that states that the HMI software will be distributed with the operating system included. The specification for the OS should include all the necessary features: networking, memory management, graphics support, security, licensing, etc. From now on, when HMI software is sold, it can be sold as a complete package. When, in the future, a computer fails after many years of service, there will be a viable path for rebuilding the same system on new hardware.

I think the most obvious response to such a demand would be a system based on BSD or Linux. Both platforms are mature and have many developers worldwide to help out. However, I also think that Microsoft could continue to dominate this area by simply licensing the source code for their older operating systems at the end of their normal support cycles.

One difficulty presented by the Windows operating platform is the rate at which the OS changes. HMI developers exhibit so much time lag while trying to release stable updates that they can easily be an entire release behind.

Either way, industrial control systems would gain more long-term stability and reliability because developers could continuously improve their own programs instead of playing catch-up to the most recent releases for a platform over which they have no control.

John Pilman
Senior Project Engineer



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Avatar replaces vehicle owner's manual

Scientists at the Technische Universitaet Muenchen in cooperation with engineers at Audi AG have developed an Avatar-based Virtual Co-driver System (AviCoS) to support a driver with explicit information on the vehicle in a natural-language dialog—supported by images and videos—making cumbersome paging through owner's manuals a thing of the past.

The avatar is displayed on the monitor of the Audi Mulitmedia Interface that comes standard in all new Audi models, reported *ScienceDaily.* The virtual figure understands complete sentences. Using artificial intelligence, AviCoS interprets questions by the vehicle occupants and answers in spoken language. The driver can view descriptive images or videos on-screen, and the avatar points to the relevant areas during the explanation.

A further option for communicating with AviCoS is a Touch&Tell mode. If a driver is unfamiliar with a specific control element, a simple touch is all it takes to cue the avatar to provide background information on the function in question. "This is a tool to explain control elements in a quick and easy, hands-on way. It is particularly useful in unfamiliar vehicles," said Professor Helmut Krcmar, chair of the TU Muenchen Institute of Business Informatics.

AviCoS can also be used while driving. To avoid distracting the driver's attention from traffic, as the vehicle speed increases, first the animations and later all graphical output is suppressed. Albeit, voice communication with the avatar remains available at all times.

"Overall, AviCoS provides comfortable

and interactive access to multimedia content that goes far beyond the information contained in printed manuals. The self-explanatory system can be used without training, making it easy to get familiar with the operation of a vehicle," said Dr. Michael Schermann, director of the Automotive Services research group at the Institute for Business Informatics.

New infrastructure propels high power AC drives market

The high power AC drives market rebounded in 2010, recovering from the economic downswing in 2009. ARC Advisory Group expects the high power AC drives market to experience strong growth during the forecast period through 2015. Emerging economies, including the BRIC (Brazil, Russia, India, and China) countries, drove growth for the high power AC drives market in 2010, and these countries also remain important markets during the forecast period.

In advanced and emerging economies, infrastructure development continued, as a significant portion of government stimulus funding was directed to that sector. Consequently, investments for high power AC drives in industries such as electric power generation and water & wastewater increased at a strong rate. Additionally, end users and OEMs across a wide range of industries have recognized that high power AC drives are a major contributor to energy savings as well as achieving sustainability.

Globalization also created a large demand for modern infrastructure, especially in emerging economies. Airport facilities and new road construction are driving demand for products from the metals & mining, cement & glass, and oil & gas industries. Emerging economies know that their current infrastructure is a huge bottleneck for their continuing high economic growth. High power AC drives will benefit in this environment as they are key components for any infrastructure development and operation.

—News brief courtesy of *Automation.com.*

Rockwell Automation starts Biofuels Customer Advisory Council

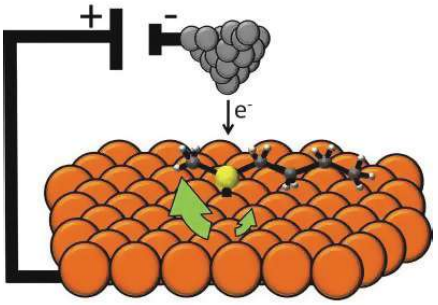
The Biofuels Customer Advisory Council (CAC) will provide customers with a forum to discuss their needs, challenges and industry opportunities. Rockwell Automation will utilize the customer feedback to guide and validate future strategies and product roadmaps for its Pavilion8 model predictive control and plant-wide optimization solutions.

The 2011-2012 CAC is composed of representatives from a diverse group of industry-leading companies, including Cardinal Ethanol, LLC, Golden Grain Energy, LLC, Kansas Ethanol, LLC, Marquis Energy, LLC, Trenton Agri Products, LLC, Western Plains Energy, LLC, Western Wisconsin Energy, LLC, White Energy, LLC and Zilor Enterprises. As part of their participation, members will have the opportunity to preview new products and take advantage of beta-testing opportunities.

—News brief courtesy of *Automation.com.*

Automation by the Numbers

1 Chemists from Tufts University's School of Arts and Sciences have created the world's first single molecule electric motor. The tiny electric motor measures **1 nanometer**, which shatters the current world record of 200 nanometers for the smallest electric motor. The team was able to control the motor with electricity through the use of a low-temperature scanning tunneling microscope. There are only about 100 of these microscopes in the U.S., and it uses electrons instead of light to identify molecules. With a metal tip on the microscope, an electrical charge was provided to a butyl methyl sulfide molecule that was positioned on conductive copper. The molecule then had carbon and hydrogen atoms radiating off of it with four carbons on one side and one on the other. These carbon chains had the ability to rotate around the "sulfur-copper bond." The researchers were able to control the rotation of the molecule by adjusting the temperature. They found that temperatures at about 5 Kelvin (K), or minus 450 degrees Fahrenheit, the motor's motion was easy to track.



The microscope sent an electrical current through the sulfur-based molecule (yellow), which was set on a conductive copper surface (orange) where carbon and hydrogen atoms radiated off of it (grey). Controlling the temperature allowed the researchers to impact the direction and rotational speed of the molecular motor. (Source: E. Charles H. Sykes)

300–400

A new creation, known as Adaptiv technology, is a camouflage cloak that masks a vehicle's infrared signature by imitating the temperature of its surroundings. BAE Systems, a British multinational defense, security and aerospace company in London, is the creator of the camouflage cloak, reported *DailyTech*. Using hexagonal panels, or pixels, which are made of a material that can change temperature rapidly, BAE Systems was able to make a cloak that not only allows tanks to mimic its surrounding temperatures, but also makes the tanks look like other objects. The hexagonal panels are operated by onboard thermal cameras, which repeatedly image the surrounding ambient temperature of the tank. The panels then project these temperatures whether the tank is moving or sitting still. In field tests, this cloaking system made a tank look like its surroundings from a distance of **300–400 meters**. To make the tank look like other objects such as cars, large rocks, trucks, etc., BAE Systems refer to a library of the heat images of these objects, and projects them onto the panels.

A heat scope detects a car, even though it is actually a tank using the heat masking technology. (Source: bbc.co.uk)

Source: BMW

1,000

BMW is working on laser-powered headlights that could debut in vehicles "within a few years," the German automaker said. The laser diodes powering the next generation of headlights will have an intensity that is **1,000** times greater than conventional light emitting diode, or LED, technology but consume only half the energy. Laser diodes will also be about 100 times smaller than the small, square-shaped LED cells, reported *TechNewsDaily*. The light from the laser diodes is blue but will be converted into a pure white light that is "suitable for use in road traffic," BMW said. The laser headlights are expected to make their first appearance in the BMW concept vehicle, the BMW i8.

61 The U.S. Steel Co. is converting its vehicles to run on natural gas, according to *Manufacturing.Net*. The steelmaker saves **61 cents** for every mile driven using natural gas instead of gasoline or diesel fuel, according to U.S. Steel Chief Executive John Surma. So far U.S. Steel has converted five vehicles at its Irvin, Pa., plant, and more are planned. The conversions cost about $12,000 to $15,000, company officials said. Chesapeake Energy Chairman and Chief Executive Aubrey McClendon said it has even bigger goals—converting its fleet of 4,900 trucks and other vehicles to run on natural gas. The company expects that will save millions each year when the process is completed in 2013 or 2014. Chesapeake is also investing $1 million toward building 1,000 to 1,250 natural gas stations across the country, McClendon said.

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Happy under pressure

Annual salary survey indicates higher job satisfaction with little or no pay increase

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By Rick Zabel

Time flies. It is hard to believe another year has passed since we published the 2010 salary survey results. In last year's salary survey article, I wrote about how I and many other automation professionals just fell into this industry at some point in our careers, and we are still here today. According to the survey results, not much has changed in a year. After all, as engineers (myself included), we do not typically like change. Change is often difficult to control, and we thrive on being in control. As a matter of fact, we get paid to be in control. So maybe it is fitting that we all work in a very conservative industry where very little changes from year to year. We appreciate the status quo.

Unfortunately, there are many factors beyond our control. In the current economic, global, and competitive environment, manufacturing companies are being forced to continually analyze, optimize, and improve production efficiencies. According to our survey, one thing does continue to change. Many of us feel greater pressure to increase productivity and reduce costs. On top of that, many of our manufacturing plants and processes are being controlled by old systems and are (or soon will be) in need of replacement or upgrades. Fortunately for us, automation plays a huge role. There are a wealth of technologies, tools, and software available to help us to address these pressures.

As pressures increase, however, the average salaries of automation professionals have not increased. Yet, we remain satisfied. Job satisfaction has increased among automation profes-

sionals. More than 80% of respondents, up 7% over last year, indicated they are satisfied with their job. 34% said they are very satisfied, and 46% said they were somewhat satisfied.

Based on the survey results, job satisfaction is tied to a number of factors. While salary is a leading factor, it is not the most important factor. Like last year, the feeling of accomplishment rated the highest, with job security, benefits, salary, technical challenge, pleasant work environment, and good relationship with work colleagues all as contributing factors. The top four most important benefits include health insurance (77%), pension plan/401K (52%), flexible working hours (37%), and paid time off (36%).

Year after year, I argue salaries of automation professionals are low when compared to the value automation brings to a manufacturing company. And each year, I keep waiting for those salaries to increase. As the workforce shortage continues to grow, I maintain the demand for experienced automation professionals will increase. I still think that time is coming. But that will require a change—one change that we engineers would certainly welcome. By the looks of it, we will still be here. After all, we are problem solvers, and what is most important to us is the feeling of accomplishment.

This year, *InTech* again collaborated with *Automation.com* to conduct the annual salary survey. The survey had 4,737 completed responses from automation professionals located around the world, with 58% from the U.S. Salaries around the world vary greatly. Instead of tallying all the results together, we decided to break out the U.S. respondents, to avoid skewing results. All the results quoted in this article, other than "Average salary by region of the world," represent U.S. responses only.

Snap shot of typical respondents

The job function of the typical survey respondent was an Automation/Control Engineer, accounting for 22% of responses. The most prominent average age range was 45–54. Nearly half (44%) of the respondents were college graduates with a bachelor's degree, with the largest percentage of those (36%) possessing a bachelor's degree in Electrical Engineering. 25% of respondents attended graduate school, and 18% received an advanced degree, of which the largest percentage (26%) acquired a Business Administration degree. The largest percentage of respondents (27%) have more than 31 years of professional work experience, while more than half (57%) have been with their current employer for less than 10 years. Salary increases were pretty much non-existent again in 2010; 25% had no increase, and 46% received a 1–3% increase. At least a portion of the compensation of 59% of our respondents came in the form of commissions or bonuses. The largest percentage of respondents (35%) clocked between 41 and 45 hours per week, and the average vacation time was three weeks per year. Let's take a closer look at the data.

FAST FORWARD

- Many of us feel greater pressure to increase productivity and reduce costs.
- More than 80% of respondents indicated they are satisfied with their job.
- The largest percentage of respondents (23%) reported a salary in the $100,000–$124,999 pay range.

Salary facts

The largest percentage of respondents (23%) reported a salary in the $100,000–$124,999 pay range. The second largest percentage (13%) was a pay range of $70,000–$79,000. The average salary in the U.S. is $99,540—that is only $203 dollars more than the average salary last year. Two other regions of the world reported a higher average salary. Canadian respondents reported an average salary of $101,646, while Australia and New Zealand respondents reported a whopping $121,089. It is interesting to note the dollar exchange rate in both of these regions is fairly close to a 1:1 ratio. However, the cost of living is typically higher in both regions, and the average salaries reflect those higher living expenses.

Average salary by region of the world

Region of world	Average salary	Percent respondents
U.S.	$99,540	57.5%
Canada	$101,646	8.2%
Mexico	$45,833	1.2%
Central America (including Caribbean)	$60,147	0.8%
South America	$62,526	4.5%
Europe (Western)	$90,196	5.0%
Europe (Eastern)	$49,479	1.1%
Africa	$63,846	1.5%
Middle East	$73,896	5.3%
Australia and New Zealand	$121,089	1.4%
Asia and South Pacific	$47,290	9.1%
South Asia	$44,673	4.4%

The average salary of the largest percentage of respondents by job function (22%, Automation/Control Engineering) was $102,660. The top five highest paid job functions are listed below—it is no surprise three of them are management positions. The highest paid job function is Consulting Engineering. With the increasing

skills shortage, this proves the demand is high for seasoned industry consultants.

- Consulting Engineering – $168,692 (3.5%)
- Engineering Management – $125,993 (5.5%)
- Safety Systems Engineering – $120,714 (0.6%)
- General or Operations Management – $114,448 (3.1%)
- Project Management – $111,060 (3.7%)

Average salary by job function

Job function	Average salary	Percent respondents
Application Engineering	$87,000	2.4%
Automation/Control Engineering	$102,660	22.0%
Consulting Engineering	$168,692	3.5%
Design Engineering	$94,318	6.2%
Engineering (Other)	$98,953	3.0%
Engineering Management	$125,993	5.5%
Environmental Controls	$85,893	0.6%
Facilities Management	$90,833	0.7%
General or Operations Management	$114,448	3.1%
Information Technology	$90,543	0.9%
Instrumentation Engineering	$105,052	5.7%
Marketing & Public Relations	$100,000	1.8%
Networking/Communication Systems	$105,313	0.3%
OEM Product/Systems Engineering	$78,864	0.4%
Operations and Maintenance	$86,718	9.1%
Plant Engineering	$98,802	1.9%
Process Engineering	$107,679	1.1%
Production Design Engineering	$105,833	0.8%
Production/Manufacturing Engineering	$76,300	1.0%
Project Management	$111,060	3.7%
Quality Control, Evaluation & Testing	$75,455	0.9%
Research & Development	$104,527	1.5%
Safety Systems Engineering	$120,714	0.6%
Sales (Inside)	$55,385	1.0%
Sales (Outside)	$107,450	6.1%
Software Engineering	$101,696	1.1%
Systems Integrator	$91,833	2.4%
Technical/Application Support	$84,432	5.3%
Training/Education	$83,333	1.3%
Other	$96,517	6.0%

A degree of higher learning

More than 69% of respondents possessed a college degree or higher. The average salary of college graduates (without advanced degrees) is $103,961. The results show those who attended at least some graduate school (but did not finish) only commanded a marginal increase in salary of $795. Those respondents who actually completed an advanced degree reported an average salary of $110, 810—that is a $7,000 increase over college graduates. If you factor that increase over your career, it certainly pays to finish that advanced degree.

The largest percentage of respondents (32%) received a bachelor's degree in Electrical Engineering, pulling an average salary of $109,492. The other top five average salaries by degree are:

- Chemical Engineering – $120,845 (10.8%)
- Physics – $112,069 (1.5%)
- Other Science – $108,136 (3.0%)
- Other Engineering – $107,173 (4.3%)
- Business Administration – $105,657 (5.1%)

Participants of our survey work in 40 different industry segments. The largest number of responses came from the Engineering Services segment (12%), where the average salary is $113,074. It is interesting to note which industries are the biggest payers. The highest average salary ($126,786) is being paid to those who work in Utilities – Pipelines, except Natural Gas. The next five highest salaries are being paid to professionals in these industry segments:

- Oil & Gas Extraction – $120,765 (4.0%)
- Petroleum Refining & Related Industries – $117,255 (6.3%)
- Engineering Services – $113,074 (12.2%)
- Utilities – Combo (Nuclear/Fossil Fuel, etc.) – $109,500 (1.6%)
- Valves, Fittings, Fabricated Metal Products – $108,750 (0.7%)

For a complete list of salary breakdowns by job function, degree, country, industry, etc., please visit: www.automation.com/salary_survey_2011.

Membership has its privileges

Namely, increased pay. Once again this year, it is apparent professionals who are members of some industry organizations pulled in higher salaries, on average, than those who are not members. Nearly half of all respondents (43%) were ISA members, and their average salary is $103,965. Compare that to respondents who don't belong to any organizations (32%), whose average salary is $92,560. Interested in ISA membership now? Visit www.isa.org/join to learn more.

Points of interest

The average salary for a male is $100,422, while the average salary for a female is $87,351—a $13,000 difference. If you go to any industry event, it is apparent a small percentage of women (less than 7% of respondents) work in the automation industry. You could argue the reason might, in part, be due to the salary gap between men and women. However, according to an April 2010 *Time.com* article, "Why Do Women Still Earn Less Than Men," a woman's salary can vary between 81% and 91% of a man's salary, when factoring comparable education and experience. So, even though the salaries are not equitable, the 87% difference reflected in our survey is within the norm. I am not saying it is right, I am just stating the facts. Arguably, one reason for the lower woman's salary is women are typically the primary caregiver for children in a family, and they are more likely to take time from their career to raise their children.

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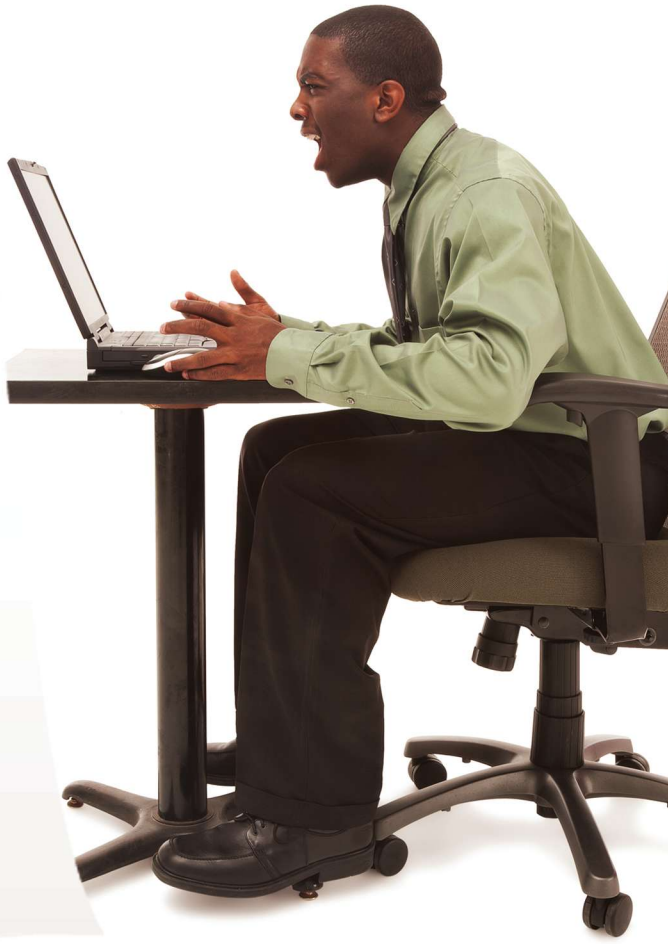

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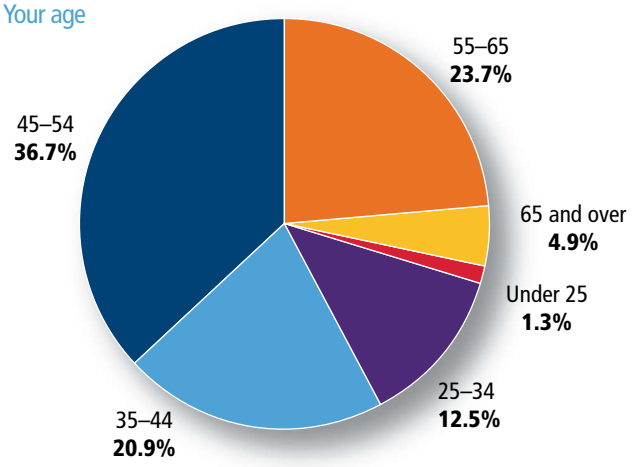
The average salary of an independent contractor was almost identical to that of a direct employee, less by $13 per year. In previous years of the survey, independent contractors typically received a few thousand dollars more per year. 93% of respondents indicated they are a direct employee, while 7% indicated they are a contract employee.

The average salary consistently increases as your tenure with a company increases. The average salary of a professional who has been at their company for less than two years is $86,731. That compares to $108,685 for those who have been at their company for 21 or more years. The average salary also consistently increases as the size of the company/division increases. The average salary of a professional who works for a company with less than 30 employees is $86,267. That compares to $113,050 for those who work for a company or division with 10,000 or more employees. The bottom line: It appears you can earn more money by working for a larger company and staying with that company for the duration of your career.

Upcoming retirements and skills shortage

There continues to be concern about the exodus of retirees from the automation industry and the imminent workforce skills shortage. More than 65% of respondents are over the age of 45 and in the second half of their career. Brace yourself for this: More than 30% of respondents will retire in the next 10 years, and almost 14% will retire in the next five years. Another 9% are not sure when they will retire. Are your companies preparing themselves to retain, replace, or augment this aging workforce?

The good news is 49% of those respondents who said they will retire in the next 10 years indicate they will continue to work part-time or offer consulting services after retirement. And who wouldn't when a "Consulting Engineer" can command a 60+% pay increase above other job functions? Another 29% of those future retirees are not sure what they will do after retirement. So, it appears there will still be a significant talent pool available, at least in the short-term future.

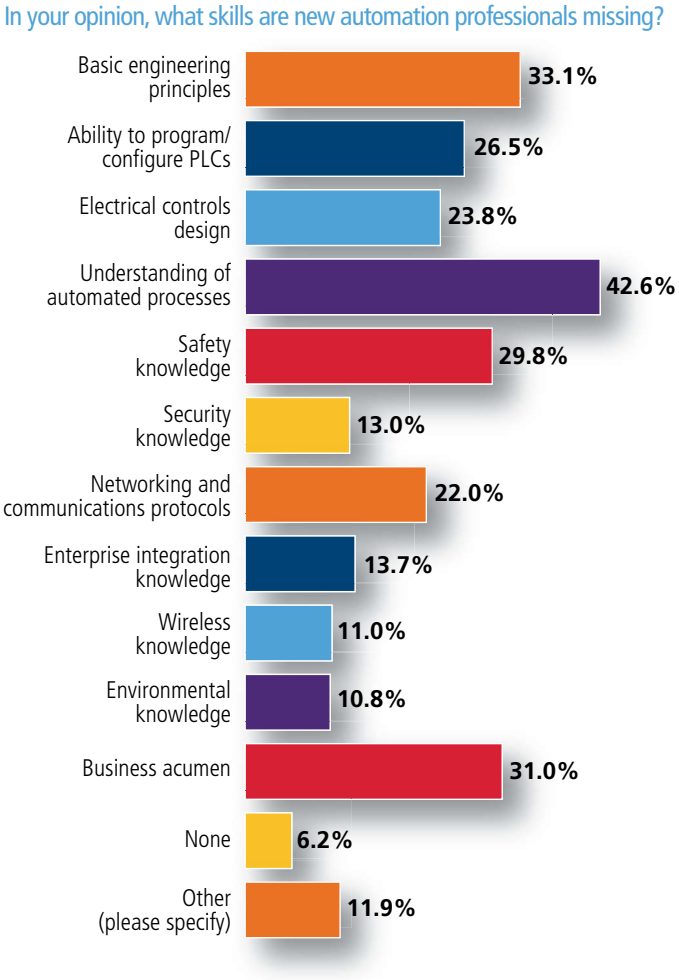


Automation newbies must brush up on skills

Unfortunately, many of those few, young professionals joining the automation industry are missing some key skills. This is not surprising because most of them did not specifically train for this industry. There are few college programs that focus on automation and control engineering, so newbies are forced to learn on the job. The top skill missing with new automation professionals is the understanding of automated processes; 43% of survey respondents selected this skill as missing. The second most lacked skill, coming in at 33%, is basic engineering principles. In third place, 31% say newbies are missing business acumen.

Pressures to reduce cost, improve productivity increase

In 2009, 81% of respondents indicated they felt an increased pressure to reduce costs. Last year, 64% said they felt increased pressure over the last 12 months. This year, 56% feel increased pressure yet again. For the majority of the remaining respondents (42%), the pressure has stayed the same. Nearly identical to the last two years, 65% percent of respondents said they felt more pressure to increase productivity, while 34% said the pressure stayed the same. While workplace pressures continue to increase, salaries do not. What is wrong with this picture?



Cooperation between Engineering & IT

Again this year, we asked the question about convergence or cooperation of IT and automation groups within companies. The complaint of dueling IT and engineering departments is continuing to subside, as it should. With the pressures to increase productivity and reduce costs, it has become essential that business processes and manufacturing process work together nicely. Up 7% from last year, 54% of respondents said their IT and engineering groups operate separately, but they cooperate well. Showing a slight decrease, 17% of respondents said their IT and engineering groups are separate, and they do not cooperate.

A few companies have even combined their IT and engineering groups. 4% of companies have combined the groups, and everyone reports to an engineering leader, while 2% have combined, and everyone reports to an IT leader. The remaining 23% of respondents said this question did not apply to them, likely because their company is too small to have defined groups or they work for a systems integration or engineering services company.

Conclusion

Not much has changed in salaries in the last year. However, based on the data from the salary survey, it is easy to conclude automation professionals are generally satisfied with their jobs. The pressure to increase productivity and reduce costs is a natural result of the tough economy and global competition. As automation professionals, we need to step up to the challenges and solve those problems that hinder our manufacturing processes and bottom-line profitability. Let's use our expertise to analyze, optimize, and improve operations. But do not forget to measure the results of your improvements. By measuring results and return on investments, you can prove to management the value of your profession and ultimately be able to command higher salaries. It is a win-win for everyone involved. This is one change you can drive.

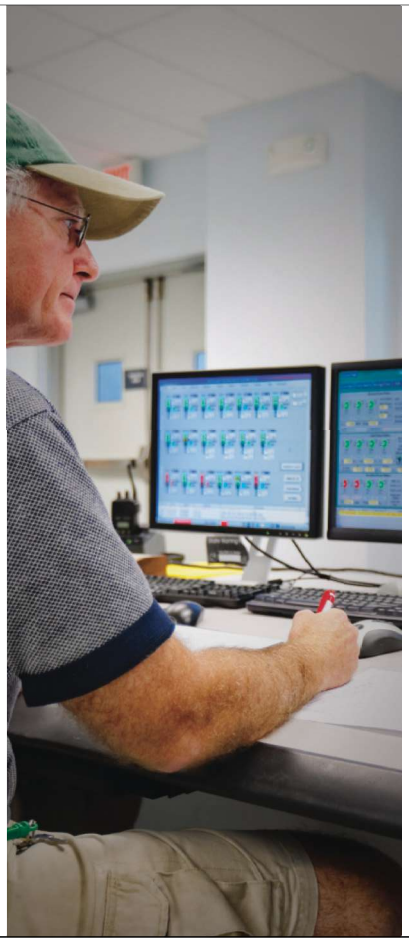
More analysis of the salary data

Because many of you are highly analytical, we published a number of salary-related tables, charts, and graphs on *Automation.com*, including a breakdown of average salaries by regions within the U.S. There are literally hundreds of ways to analyze and compare the data, but we had to stop somewhere. Go to www.automation.com/salary_survey_2011, and see how you stack up.

ABOUT THE AUTHOR

Rick Zabel is vice president and publisher of *Automation.com*. He would like to extend a special "Thank You" to all who took the time to complete our survey; and to Kia Weller and Stephanie Dwyer at *Automation.com* for all their help compiling the survey data.

View the online version at www.isa.org/intech/20111001.



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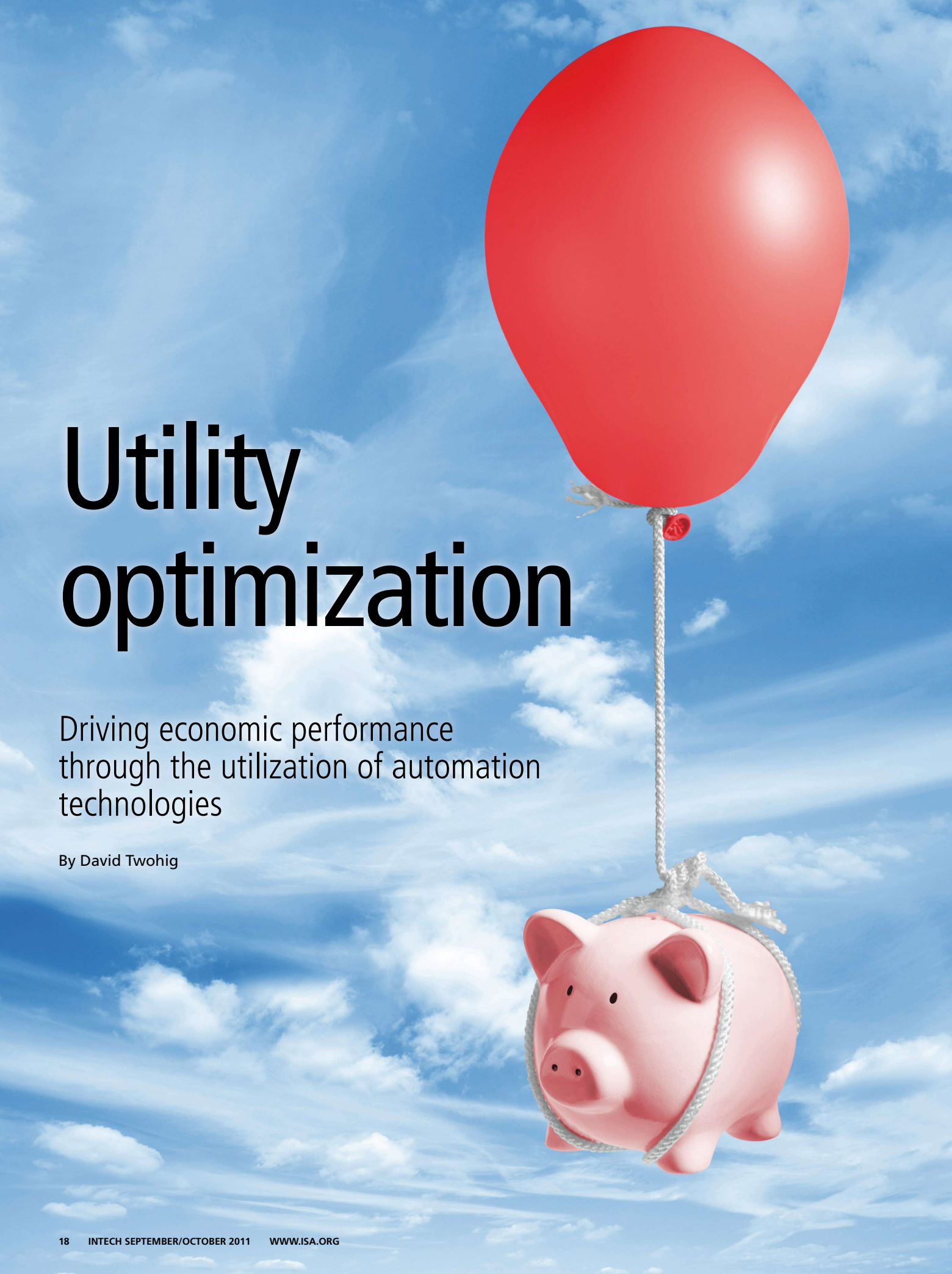


Utility optimization

Driving economic performance through the utilization of automation technologies

By David Twohig

Improving business performance comes as a pre-requisite when you work as an engineer in the manufacturing industry. While this can be achieved in numerous ways, cost reduction is the obvious primary objective for most companies. Operations are continually trying to achieve greater throughput for less input. Therefore, this has become the long-standing challenge to automation systems, as the business tries to become more competitive in expanding their market share and gaining that competitive edge. It is no longer acceptable for control systems to just manufacture product or manage utilities effectively; we must maintain those standards but achieve them more efficiently.

In practical terms, when you consider the different ways of trying to achieve these objectives, improving your energy efficiency is an obvious choice. This case study outlines the benefits of how various automation technologies were utilized to reduce costs and improve the performance of a large compressed air system.

Background

Typically, two of three air compressors are used to supply aeration to an Active Pharmaceutical Ingredient fermentation process. The compressors draw in air from the local atmosphere, and in compressing the air, supply a common mechanical header. Individual fermenter vessels then manipulate their own aeration demand from this header through an air flow control loop. The principle of operation for the compressors originally consisted of the production team entering a common fixed pressure setpoint through the SCADA HMI. The compressors then worked to maintain this setpoint by manipulating their volumetric throughput. Presently, the system is configured to run one compressor at full capacity while the secondary compressor varies its supply as the process demand for aeration oscillates. (Note: This configuration is applied due to the age of the equipment. The older, less efficient compressor is run at its optimal point, which is approx 100%, while the secondary compressor is utilized to vary its output because it has a better efficiency curve.)

Because the supply of aeration is critical to the manufacturing process, historically these compressors were manually manipulated to ensure there was an excess of air in the system. This practice also served the purpose of ensuring the compressors were operating safely within their own performance curves, i.e., away from surge. (Note: Surge is defined as the point at which the compressor cannot add enough energy to overcome the system resistance or in this case the header pressure. This causes a rapid reversal of flow, or surge, back through the system, which can result in vibration and mechanical damage.)

This excess air was then vented to atmosphere via a blow-off valve or through a spare fermenter vessel. And although this method was inefficient in terms of energy usage, it did provide the production team with an aeration buffer. This buffer was used to absorb sudden process oscillations that might impact product quality. And so, over the 50-year life span of the system, this technique became culturally ingrained in the team when running the compressors, and as a result, the systems' overall efficiency suffered, resulting in high running costs.

Historically, this culture would not have been perceived to be a problem given the cost of manufacturing product dwarfed the associated energy costs; however, the company has since become increasingly focused on the unit cost of energy. In addition, people have become more environmentally conscious, and so the expectation is to become more energy efficient. While previous projects had attempted to optimize the performance of the compressors, it proved difficult due to the limited availability of process data from the compressor control system. The net result of all of the above was we knew the system needed improving, but quantifying and evaluating success with its current configuration was going to be difficult. However, because the cost of energy has spiralled, and the culture in managing the compressors was poor (i.e., there was obvious wastage), the system did present itself as the proverbial "low hanging fruit" with regards to

FAST FORWARD

- Improving business performance has always been an expectation of engineers working within the manufacturing industry, never more so than the last number of years.
- This case study details how a number of different automation technologies and techniques were utilized to identify performance issues and deliver cost-saving solutions on a large fermentation compressed air system.
- Simple, robust, low-cost automation solutions used to deliver improved business performance through energy reduction and operational excellence—8% reduction in annual electrical charge and 3% reduction in CO_2 targets.

In practical terms, when you consider the different ways of trying to achieve these objectives, improving your energy efficiency is an obvious choice.

reducing manufacturing costs, and so we were about to try again. After some initial investigating, it became apparent the automation technologies needed to drive improvement had evolved over the last number of years and become more reliable and affordable. Therefore, the opportunity to drive improvement became more of a realization.

To ensure the potential savings were correctly identified and quantified for the business, a Lean Six Sigma project was implemented to quantify, optimize, and validate all elements of improvement technically and culturally.

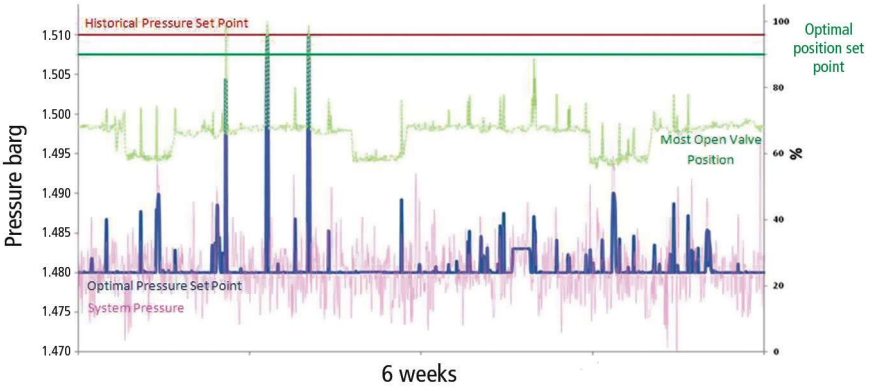
System overview

Although the fermentation process relied on the supply of air, the compressors were controlled by an independent system that did not communicate with the main Fermentation DCS. The compressors were controlled on a PLC/SCADA system that operated on a standalone network. By contrast, the fermentation process was controlled by a large DCS that was fully integrated into the site historian. And even though they operated as systems in isolation, they were heavily dependent on each other to ensure manufacturing ran smoothly (i.e., fermentation depended on the supply of aeration, and the compressors depended on fermentation to vary its demand proportionally to ensure they avoided surge conditions). As a result, the production became the common denominator in this balancing act.

Project life cycle

To ensure the project started effectively, different automation technologies were utilized to interface the compressed air control system with the site historian and Fermentation DCS. From this integration, it became possible to evaluate the performance of the compressors in terms of their energy consumption and process performance, particularly with respect to fermentations' processing activities.

By correlating this data with the data from the electrical energy meters on each compressor, the project team was able to quickly identify and quantify the performance/cost gaps. Utilizing this data further, the team was able to develop individual energy profiles for each compressor, and these profiles provided the metrics to which the project would validate its success.

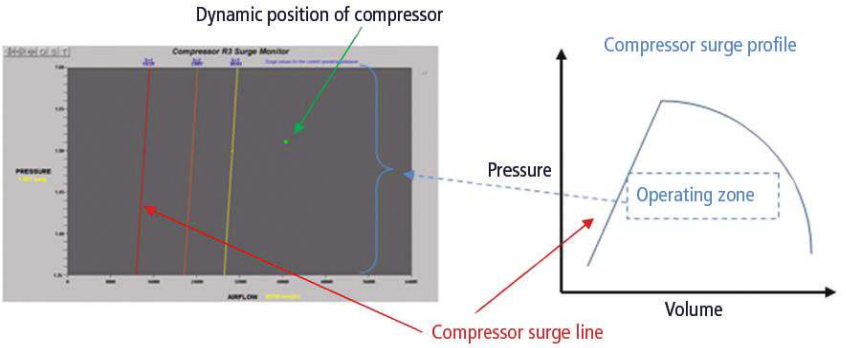


Having examined the overall performance of the entire system, there were two primary issues with the supply of aeration. First, the compressors were utilized to produce an excess of air. Over certain periods, this could be as high as 15%, and in producing this air they did so at a pressure higher than was required. Referring to engineering first principles and heuristic data, the team was able to conclude the power utilization of the compressors was proportional to the volume of air being produced and the pressure at which it was produced. Therefore, the solutions appeared obvious—eliminate the waste air and optimize the pressure. Simple in theory! However, in reality, the techniques to enable these solutions needed to be identified and then implemented on a live manufacturing process.

… the solutions appeared obvious—eliminate the waste air and optimize the pressure. Simple in theory! However, in reality, the techniques to enable these solutions needed to be identified and then implemented on a live manufacturing process.

Initially, a solution was put forward based on ISA's *Instrument Engineers' Handbook, Third Edition* – Section 8.9 Optimized Load Following whereby the positions of the air-flow controllers on the supply line of the fermenter vessels could be used to determine the optimal operating pressure of the header system. By using a simple selector block to determine which valve across all the fermenter vessels was in the most open position, that value was then compared to an optimal position of 90%. Through an integral-only controller, the difference in this positional relationship was used as the offset to determine the optimal pressure setpoint for the system. (Example: As the pressure setpoint reduced, the varying compressor would reduce its volumetric throughput. As this reduced the volume of air in the header, the supply valves of the fermenter vessels would open up to maintain their recipe setpoint. The setpoint of the pressure controller would reduce the air in the system until the position of most open valves would reach 90%.) This simple solution utilized the valve positions of the vessels as an indirect indication of the air requirements of the fermentation process. And so, increased or decreased, the system pressure to optimize the electrical utilization of the compressors, while still ensuring sufficient aeration, was supplied.

We can see the results from the initial trial period (see chart above), where the optimized pressure setpoint, i.e.,



the purple line hugs the lower pressure limit of 1.48 barg except for short infrequent periods of high demand. (Note: This data is provided from the initial six-week commissioning period in which a lower limit of 1.48 barg was applied.) However, if you examine the secondary axis (green lines), you will notice there is still a sizeable difference between the active most open valve position and the optimal setpoint of 90%. This was a clear indication the system was producing sufficient aeration at the lower pressure. And that the pressure setpoint could be reduced further in order to drive the most open valve to the optimal setpoint and capitalize on further energy savings.

In terms of dealing with the volumetric wastage, there were two challenges. First, there was an operational issue with regards to the production teams' understanding of how the compressors should run effectively. They knew surge conditions were to be avoided, and so increasing the volumetric throughput of the compressors via wastage provided a "comfort zone" in doing so. To improve the daily management of the compressors, the production team had indicated if they had a simple way of monitoring surge, they would be willing to operate closer to the edge of the proverbial surge cliff. In response, the automation group was able to capitalize on the integration of both systems and utilized existing compressor PLC code to develop some surge profile displays for the fermentation HMIs. Because the PLC control strategy utilized some simple mathematical techniques to develop its own compressor profile, these displays were able to convey the dynamic position of a compressor in relation to its surge profile. In simple terms, the operator could look at the screen and under-

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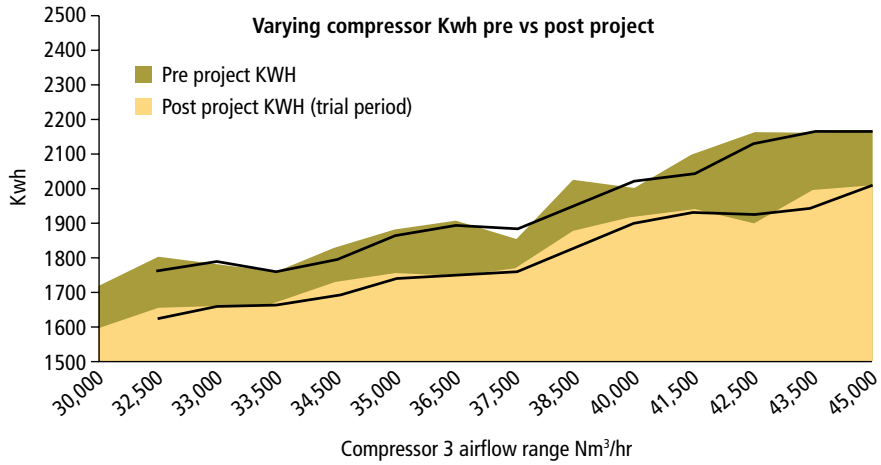

stand the compressor needed to operate with a specific envelope and as a result could pre-empt manufacturing decisions to ensure it remained within said envelope.

In the compressor charts, we can see the operating zone displayed on the operator HMI showing the dynamic position of the compressor relative to some earlier warning alarms and the actual surge line.

Second, there were legitimate situations in which some wastage would be required. One example was if the manufacturing demand for air was more than one compressor could supply but less than the minimal output required to run both compressors safely.

To address this legitimate wastage requirement (although tolerating wastage is technically not a Six Sigma policy), the automation group developed an "Air Waste Management Strategy." Implementation of this software solution enabled the production team to activate a recipe in the DCS that would continually minimize the waste air required to safely run two compressors. It did this by manipulating the aeration throughput of an empty fermenter vessel based on the difference between the real demand and the minimum throughput required to run two compressors. By creating this dummy demand, the compressors interpreted it as a real aeration requirement and so responded accordingly. This resulted in

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the compressors operating safely away from surge but had the added benefit of increasing or decreasing this dummy demand as production's requirements oscillated, i.e., as legitimate demand came online, the software recipe would reduce the air demand of the dummy vessel.

Results

Because the Six Sigma concept is to categorise results in terms of benefit, i.e., Type 1 to 4 with Type 1 being balance sheet savings and Type 4 being indirect savings such as operational improvements, it therefore provided the team with a strong framework in which to quantify the success of the automated solutions.

Initially, the team utilized compressor energy profiles to determine certain savings. In this example, we can see the impact the pressure reduction strategy had with respect to the varying compressor. Aligning the "Varying Compressor" chart with the earlier trend, and it is evident the compressor is producing sufficient air volume, but in doing so at a lower pressure, it has reduced its energy consumption. (Note: This analysis represents the previous chart, and so there is a lower pressure limit in place. Because the process data indicates the limit can be reduced further, we can conclude greater energy savings are to be realized once the positional setpoint is optimized.)

… low-cost automation solutions can have a large impact on improving business performance, and automation has a large role to play in driving cost reduction for companies.

In addition, the team was able to quantify a reduction in volumetric wastage utilizing the site historian. It was evident from the data that the air being produced was being utilized more efficiently compared to before the project. The direct savings have been quantified in the region of an 8% reduction of a typical annual electrical charge and 3% reduction in the sites' CO_2 target emissions.

While this is considerable, there have also been other significant improvements that have contributed to improved business effectiveness. The site historian is now used by the production and support groups to track the performance of the compressors. By monitoring various trends in conjunction

to the HMI displays, the production team has been able to improve the day-to-day management of the compressors, keeping costs and environmental impact minimized.

In addition, since the process data has become available, the maintenance team has been able to monitor the performance of the compressors' equipment more effectively. This monitoring has enabled the team to proactively calibrate equipment and prevent situations that historically have shown the compressors to trip out. Because of this pro-active culture, the compressors have tripped out only once in 18 months compared to the three trips in the first quarter, prior to the project being implemented. This has not only led to improved performance, but it has contributed to changing the culture of the production team with regards to managing the compressors.

Closing statement

While this is a specific example that may not be applicable to all compressed air systems, it does demonstrate how, with some lateral thinking, different automation concepts can be applied to reduce costs and limit environmental impact. When you consider that the most significant contribution in terms of reducing the costs of running the compressors was the pressure reduction strategy, this in automation terms is just a simple integral only controller. The case study therefore demonstrates that low-cost automation solutions can have a large impact on improving business performance, and automation has a large role to play in driving cost reduction for companies. The smarter use of our systems should be seen as the key enabler when it comes to improved business performance and optimizing existing utilities.

RESOURCES

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David Twohig (Twohig_William_David@Lilly.com) has been working as an automation engineer for Eli Lilly in the U.K. for a number of years. He has a degree in Applied Physics and is presently working on a research Masters with the Chemical Engineering Dept. at Newcastle University. In addition, he is a full member of the Institute of Engineering & Technology and is presently serving as an executive committee member for the Control and Automation Community.

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Vibration analysis goes mainstream

With advances in sensor, recording, and analysis technology, vibration analysis is now within the reach of even small organizations

By John Bernet

Most machines have rotating parts, and rotating parts vibrate. Measuring how and how much those parts vibrate can tell you a lot about the health of a machine. Whether it is the rumble of worn bearings or the shaking, shimmying, or thumping of loose, misaligned, or unbalanced parts, machines have a tale to tell to those who are willing and able to listen.

Vibration analysis—the art and science of measuring and interpreting those telltale rumbles and shakes—has been around for decades, but mostly in the domain of specialists operating exotic instruments for corporations and government agencies with mission-critical equipment and very deep pockets. For everyone else, "vibration analysis" was typically performed by a mechanic using a makeshift stethoscope fashioned from a screwdriver—the tip held to the machine, the handle held to the ear—or, more often, not done at all. Recent developments in vibration sensor, data acquisition, and analysis technologies, however, are making vibration analysis cheaper, easier, and more widely available.

What vibration analysis tells you

Among the most important mechanical faults that vibration analysis can reveal are:

- **Imbalance**: A "heavy spot" in a rotating component causes vibration when the unbalanced weight rotates around the machine's axis, creating a centrifugal force. As machine speed increases, the effects of imbalance become greater. Imbalance can severely reduce bearing life as well as cause undue machine vibration.
- **Misalignment/shaft runout**: Vibration can result when machine shafts are out of line. Angular misalignment occurs when the axes of (for example) a motor and pump are not parallel. When the axes are parallel but not exactly aligned, the condition is known as parallel misalignment. Misalignment may happen during assembly or develop over time, due to thermal expansion, components

Technician using a handheld vibration analyzer

shifting, or improper reassembly after maintenance. The resulting vibrations may be in the direction of the rotation, along the shaft axis, or both.

- **Wear**: As components, such as bearings, drive belts, or gears, become worn, they may cause vibration. When a roller-bearing race becomes pitted, for instance, the bearing rollers will cause a vibration each time they travel over the damaged area. A gear tooth that is heavily chipped or worn, or a drive belt that is breaking down, can also produce vibration.
- **Looseness**: Vibration that might otherwise go unnoticed may become obvious and destructive if the component vibrating has loose bearings or is loosely attached to its mounts. Such looseness may or may not be caused by the underlying vibration. Whatever its cause, looseness can allow any vibration present to cause damage, such as further bearing wear or wear and fatigue in equipment mounts and other components.

Measuring vibration

Vibration sensors have advanced far beyond the mechanic's screwdriver. There are a variety of sensor types, but the accelerometer is the most common. To take a measurement, a small metal sensor is attached to the appropriate location on the equipment to be tested. The attachment, which can be permanent for continuous monitoring or temporary for machines that are evaluated only periodically, must be at a position on the machine that reveals the best information about the vibration being investigated (at the bearings of a motor, for example, or close to a rotating shaft).

Inside the sensor, an array of tiny electronic accelerometers convert movement along any of the three axes (up and down, back and forth, side to side) into an electrical signal fed to a recording device. Recorded vibration data can be analyzed at the test site for an immediate diagnosis and can also be saved for later analysis or comparison with earlier recordings to monitor trends in machine health.

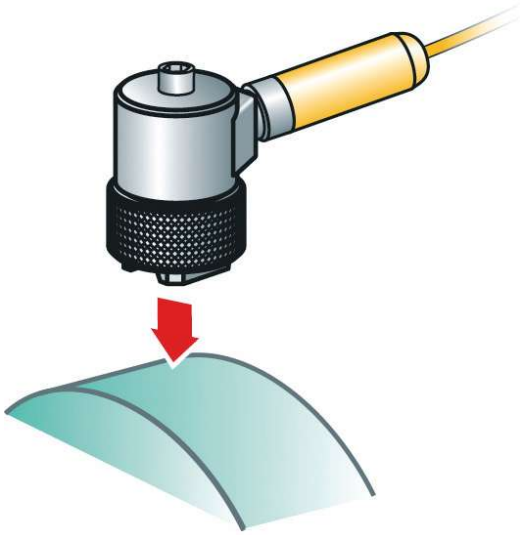
Studies conducted by the U.S. Navy found many vibration analysis programs were not collecting all of the data needed to make an accurate diagnosis. The studies concluded that to diagnose machine condition accurately, data was needed from all three axes of a rotating shaft. When only two axes of data were used, diagnostic accuracy dropped to 80%. When data from only a single axis was analyzed, diagnosis accuracy dropped to 46%.

Automating the analyst

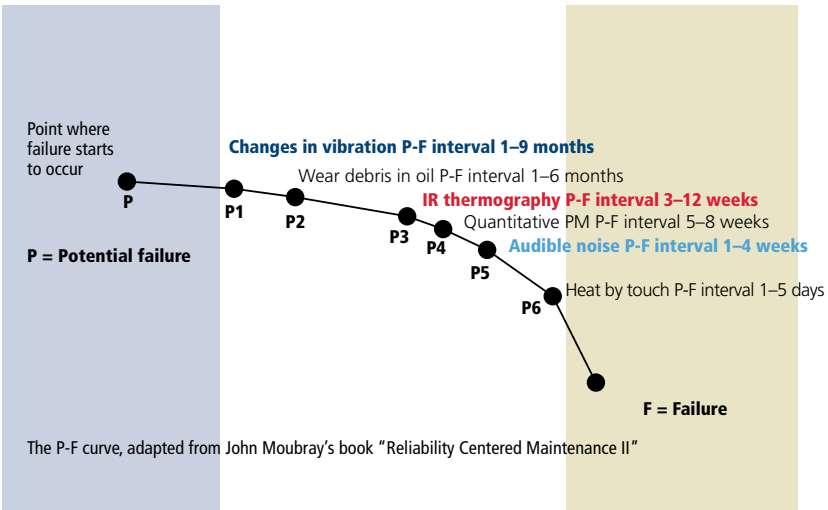
Collecting and storing vibration data from a sensor is only the beginning. To be useful, vibration data must be analyzed and interpreted. A vibration graph can reveal a lot to a trained and experienced vibration analyst, but hiring (or training and then retaining) a vibration analyst is such an expensive proposition that only large, well-funded organizations have been able to afford to keep analysts on staff. Everyone else has had to hire vibration consultants only when need is

FAST FORWARD

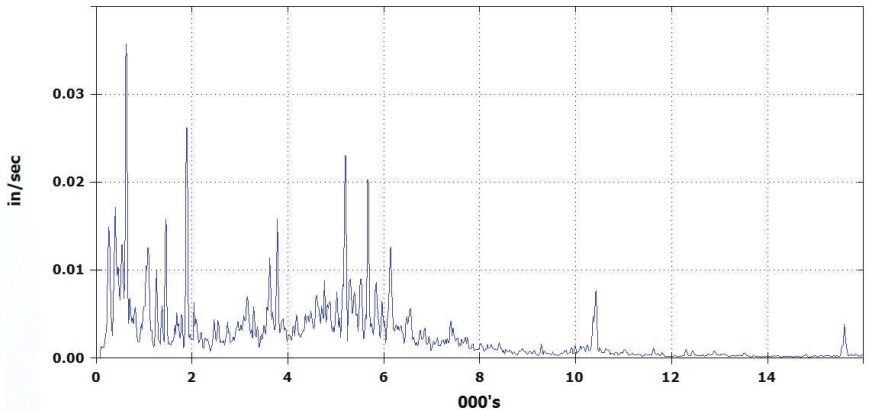
- Measuring the vibrations of motors, pumps, and other common machines can reveal valuable information about machine health—or impending failures.
- Vibration analysis can reveal four of the most common mechanical faults: imbalance, misalignment, wear, and looseness.
- Easier measurement procedures combined with automated vibration analysis enables personnel with minimal training and experience to use vibration to evaluate machine health and determine required maintenance.



Attaching a vibration sensor with a magnetic mount



Potential failure curve over a nine-month interval



Vibration intensity and frequency of a rotating shaft

justified and budget allowed.

Making vibration analysis available and affordable for everyone who could benefit from it would require not just affordable equipment but also "automating the analyst." Automated diagnostic programs were needed that could analyze raw vibration data and give useful, simple, "actionable" recommendations for non-experts. The key to automating vibration analysis, as it turned out, was to compare the vibration data in question with data from a similar, "healthy," "known good" machine. Although the concept of comparing the data from the machine in question with "baseline" data from a similar, known-good machine is simple, the implementation is complicated. A vibration analysis program performs a sophisticated analysis, comparing hundreds of data points with the "fault patterns" of similar machines to give a simple, understandable, diagnosis that makes clear how healthy the machine is and whether maintenance is needed. The resulting diagnostic report should give the operator or maintenance technician a clear picture of machine condition and action required.

Benefits of vibration analysis

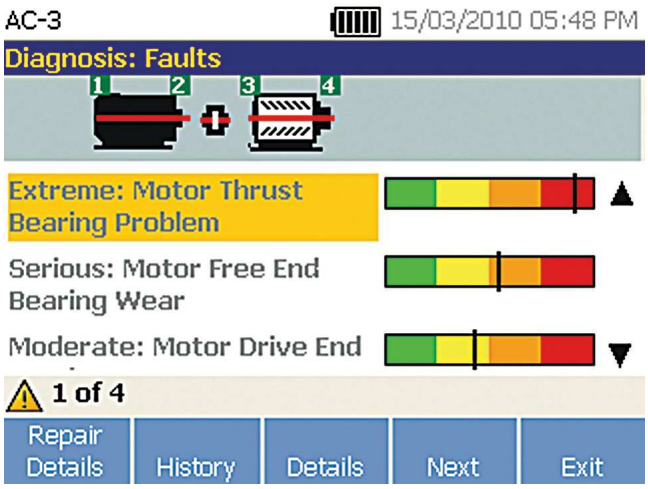
- **Predictability:** Studies have shown vibration analysis can provide early warnings of impending machine failure, giving maintenance staff time to schedule required repairs and acquire needed parts.
- **Safety:** Having information about machine health enables operators to take faulty equipment offline before a hazardous condition occurs.
- **Revenue:** Well-maintained machines have fewer unexpected and serious failures, helping to prevent production stoppages that cut into the bottom line.
- **Increased maintenance intervals**: When machine health is being tracked, maintenance can be scheduled by need, not just by accumulated hours of operation.
- **Reliability:** Monitored machinery has fewer unexpected or catastrophic failures.
- **Cost savings:** Running machinery until failure often results in more expensive repairs, overtime, and forced purchases. Twenty-five years of documented savings show a 20:1 benefit-to-cost ratio for vibration analysis programs.
- **Peace of mind:** A better understanding of machine health builds confidence in maintenance schedules, budgeting, and productivity estimates.

Why add vibration analysis to maintenance program?

"Run to fail" maintenance programs, while simple, often have costlier repairs, loss of revenue from production stoppage, and expensive overtime. Preventive maintenance programs, in which machinery is serviced after a certain number of hours of operation, can result in unnecessary work being performed, and unmonitored machinery can still fail before the maintenance interval elapses. When vibration analysis is incorporated into a maintenance program, however, the condition of monitored machines is known, so unnecessary maintenance work is avoided, and required work can be scheduled for convenient times and when parts are available. Maintenance staff knows which machines are good enough to run, which need repairs scheduled soon, and which need to be shut down before they fail.

Getting started with vibration analysis program

The U.S. Navy determined 30 years ago it wanted the benefits of vibration analysis, but could not afford to have a vibration expert on every ship. What are the roadblocks to implementing a program?



Automated vibration analysis can report on machine health in terms that are understandable and actionable by technicians without vibration analysis experience.

- Training staff and then retaining staff with the extensive vibration analysis skills is expensive.
- Results achieved may not justify the cost in equipment, training, labor, etc.
- Company priorities change, so a vibration program is scrapped.

Recent advances in vibration analysis, however, have enabled programs that can diagnose common machine faults without the need for prohibitively expensive equipment and expert operators.

The suggestions below can help any organization in its efforts to implement a vibration analysis program.

- Start small and grow. Do not try to monitor 500 machines in a plant all at once. Instead, choose 25 to 50 machines to start with, then add additional machines as priorities, time, and budget allow. Organizations that already have a reliability group can increase the scope of their maintenance program to include vibration analysis.
- Focus on problem machines. If you have machines that have a history of failure or a few machines that can take down half the plant, start with them. Even small machines that are not deemed big enough for a reliability group to monitor may be important to the maintenance and operations groups because they are the ones that require the most attention.
- Focus on the common machine faults—imbalance, misalignment, looseness, and bearing failures—because they account for 80-90% of machine faults.
- Use automation and proven measurement methodology to get a complete picture of the machine's entire power train. Maintenance technicians and operators do not have time to look over reams of data—they have a plant to run. A system that screens the data and provides answers about what is wrong with a machine and what to do to fix it should be the goal.

Summary

Advances in vibration sensor, data acquisition, and analysis technologies have enabled the introduction of powerful, portable, affordable, easy-to-use vibration measurement and analysis tools that enable even smaller organizations with limited training and hardware budgets to enjoy the considerable benefits of vibration analysis.

ABOUT THE AUTHOR

John Bernet (John.Bernet@fluke.com) is a vibration application specialist at Fluke Corporation and a Category II-certified vibration analyst. He has more than 20 years of vibration analysis experience in industry and the U.S. Navy.

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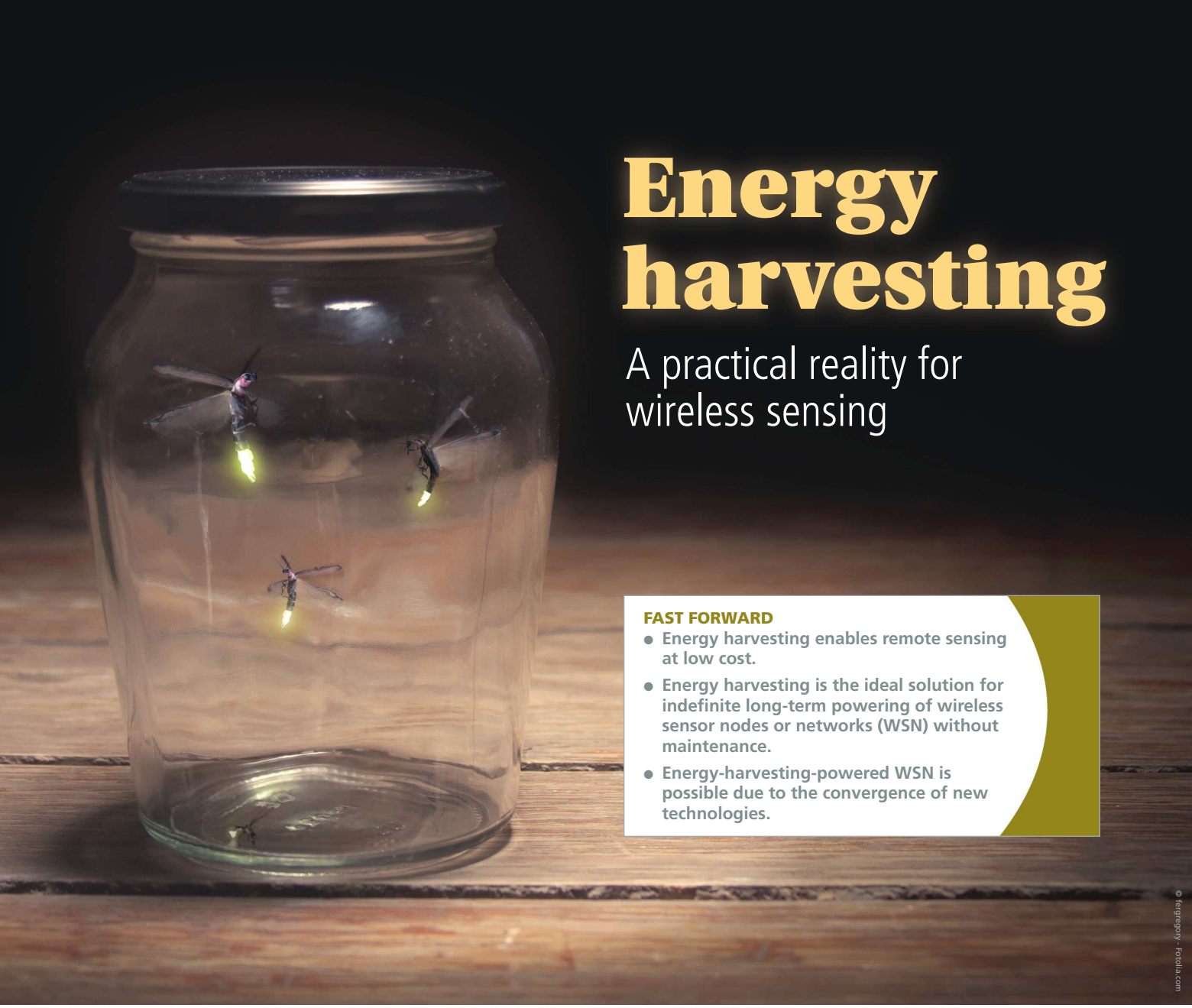

Energy harvesting

A practical reality for wireless sensing

FAST FORWARD

- Energy harvesting enables remote sensing at low cost.
- Energy harvesting is the ideal solution for indefinite long-term powering of wireless sensor nodes or networks (WSN) without maintenance.
- Energy-harvesting-powered WSN is possible due to the convergence of new technologies.

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By Roy Freeland

There are some exciting high-growth projections for wireless sensing for the automation industry. More sensors mean more process efficiency, lower operating costs, lower maintenance costs, higher reliability, and greater safety. Wireless sensing provides the opportunity to install masses of sensors with virtually no cost of installation by reducing the need for cables carrying the signals from the field to the control room. Wiring costs can easily be 80%, or more in a hazardous area, of the total cost of installing a new sensor. Who wouldn't like to get the same job for one-fifth of the cost or five times as many sensors for the budget? And it isn't just the cost of the installation; there are many cases in which a plant has to be shutdown to facilitate installation, adding another massive sum to the cost of new sensors.

Most of us routinely use wireless (cell phones, Wi-Fi) for communication, and the potential for M2M (machine to machine) wireless communication is considered to be even larger. Wireless transmission of sensor data is now well established as a reliable method of monitoring industrial plants. It is even being perceived by some users as more reliable and maintenance-free than hard wiring.

This new approach to automation has been made possible by the convergence of new technologies:

- Low-power electronics, including microprocessors with sleep modes
- RF transmission systems that use digitally encoded signals (e.g., digital television and Wi-Fi) with an order of magnitude less power required than older analog systems
- New energy harvesting techniques

So why is there so much interest in energy harvesting? Simply, you cannot get the full benefit of wireless unless the power source is also wireless. This means you need either a battery or some form of energy harvester. Until recently, the usual power source available to power a wireless sensor node or network (WSN) has been batteries. With their limited and non-deterministic lifespan, hazardous content, shipping, and disposal requirements, batteries alone are not likely to provide a power source that will last the life cycle of the WSN application without maintenance intervention. The ideal solution is an energy harvester that is "fit and forget" and will have a lifespan in excess of the WSN that it is powering.

What is energy harvesting? Energy harvesting is the extraction of usable energy (usually converted into electrical energy) from otherwise wasted energy available in the environment. On the macro scale (MegaWatts - MW) this includes hydro-electricity, wave power, solar panels, and wind turbines. However for wireless sensing, we are talking about harvesting immediately available energy such as vibration, heat, light, and RF energy to produce milliWatts - mW.

Power requirements for WSNs

Whether the power source is an energy harvester or a battery, it is important to minimize power consumption. Much can be done to minimize average power requirements, for example reducing reporting frequency. If a wireless system is being used for machinery condition monitoring, then it is unnecessary to specify the transmission of full vibration spectra every minute, when it is replacing a man on a bicycle with a hand-held device who goes around once a month (provided it is not raining and he has nothing more urgent to do). Also parameters can be monitored and analyzed in the WSN, and it can be programmed to transmit alarm signals only when there is a problem.

To illustrate the issues, this article takes the example of a WSN that requires an average power of 3mW to compare various options. This is not untypical of either a frequent reporting requirement (such as several times per minute) or a high-data requirement (such as complete vibration spectra).

The following table shows the theoretical life of standard-sized cells from a leading Lithium battery manufacturer. In practice, the theoretical capacity is reduced by such factors as the need for intermittent high currents for RF transmission, self discharge, and low temperatures. Some newer designs perform closer to theoretical capacity and may include energy storage to help with the peak power requirements of WSNs.

Battery size	Nominal capacity	Life at 3mW (3.6V)
AA	2.4 Ah	Less than 3 months
C	8.5 Ah	Less than 10 months
D	19 Ah	Less than 2 years

Energy harvester power

So what are the options for energy harvesters to deliver 3mW? The following are systems that are available today, and they represent each of the main types of energy source that can be used in practice in many types of plant and other machine applications to provide the required power. Each of these uses a source of energy that is readily available in many but not all applications; however with this choice, it should be possible to select a suitable device for the vast majority of applications.

1. **Vibration**: Perpetuum's vibration harvester will produce 3mW from about 40-50mg of vibration, depending on the exact frequency. Its bandwidth is important to ensure adequate coverage of a wide range of machines.
2. **Heat**: Micropelt's thermal harvester will produce 3mW from a suitable heat source at about 75°C assuming ambient temperature of 25°C. The rate of heat transfer is important and installing a probe in hot liquid flow reduces the temperature required for 3mW to 55°C.
3. **Photovoltaic**: G24 Innovations Photovoltaic Dye sensitized thin film photovoltaics require an area of 233mm x 135mm to produce 3mW in a typical industrial indoor environment with a light level of 500 lux.
4. **RF power transmission**: Powercast's RF transmission system requires a 3W transmitter to

Power source standards

The ISA100.18 Working Group is preparing standards and information documents on power sources for WSNs. Key objectives are to define specifications for the interchangeability of various power sources, including batteries, energy harvesters, and other possible types, such as 4-20mA loops, and to define performance specifications so users can compare different harvesters and choose the optimum power source for each application. The working group is cooperating with a range of organizations, including VDI and NAMUR on battery standards for WSNs and other organizations using 802.15.4, such as WirelessHART and Zigbee as well as other low power wireless protocols.

How to calculate the power needed? Typical wireless sensor nodes have a duty cycle with varying power requirements ranging from "sleep" or "quiescent" modes, where little is happening and power consumption may be of the order of 0.1mW or less, to brief bursts of higher consumption when microprocessors are handling and interpreting data with peaks of power of 100mW or more when the RF transmission occurs. The energy harvester does not normally supply the peak level of power continuously but charges up a capacitor, supercapacitor, or rechargeable battery to provide the peak-power requirements. The important calculation is, therefore, the average power required over the complete duty cycle, including inactive periods.

produce 3mW of usable power at a range of 1.2M (4ft). This system is technically wireless power transmission rather than energy harvesting.

Practical applications

End users who have trialed battery powered WSNs have generally become very enthusiastic about the benefits. However, we are now seeing views being expressed that the power supply issue must be resolved and that changing batteries is not acceptable in most industrial situations. This is not only because of the cost of the work to order, stock, organize, and physically replace batteries, but also, particularly in hazardous area and inaccessible areas, there is an understandable reluctance to send maintenance staff into potentially dangerous areas. The major systems builders are, therefore, almost without exception working on offering energy-harvesting-powered options for their WSNs.

A good example is the GE Bently Nevada wireless condition monitoring system installed as a pilot at Shell's Nyhamna Gas Plant for predictive maintenance. This was a site built in Norway to process natural gas from the Ormen Lange field in the North Sea and pump it across to England. Although it was a greenfield site, it was found that the cost of hard wiring was excessive to monitor most of the plant. Therefore a wireless system powered by vibration energy harvesters was used on a number of machines to provide full vibration data from accelerometers to the central data processing system.

Newer installations with the latest vibration energy-harvester-powered system have been installed in power stations. It is notable that previous experience with the power available from vibration harvesters leads to a decision to use one harvester to power a node with four sensors rather than the previous ratio of 1 to 1 for harvesters and sensors. It is a fascinating insight into the business case that it is economical to use vibration energy harvesters to produce milliWatts of power in a plant that is producing Megawatts.

Micropelt's thermal harvester is being used to monitor the temperature of power busbars to identify critical situations. Any rapid rise in temperature will cause an alarm to be transmitted wirelessly to a control room.

National Instruments recently started to offer vibration and photovoltaic solutions to powering their wireless devices.

The future

The benefits of using wireless for automation monitoring and eventually control are so strong that practical solutions for suitable power sources will continue to develop. Energy harvesting has many different forms that have been fully demonstrated to be ideal solutions for indefinite long-term powering of WSNs without maintenance. Although the power requirements of some electronics will continue to fall, we are probably getting close to the limit of low-power RF transmissions as well as the chemical energy density possible in primary battery cells. The energy available from various energy harvesting techniques in most applications already significantly exceeds the power requirements of existing WSNs. The recent rush to design in energy-harvesting options for battery-powered WSNs will not only lead to much wider use of energy harvesters but also ensure much wider use of low-cost wireless sensing with all the benefits of increased monitoring for plant safety and efficiency.

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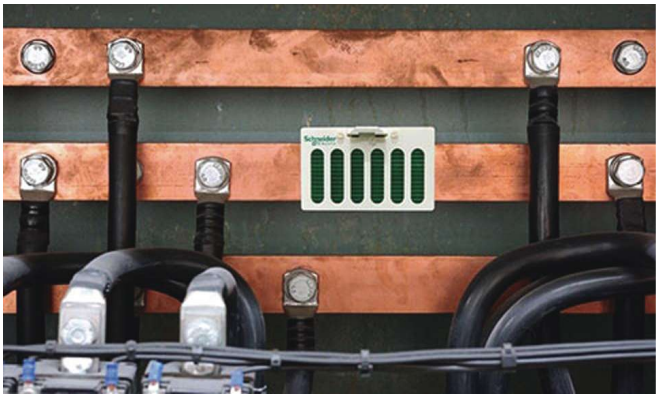
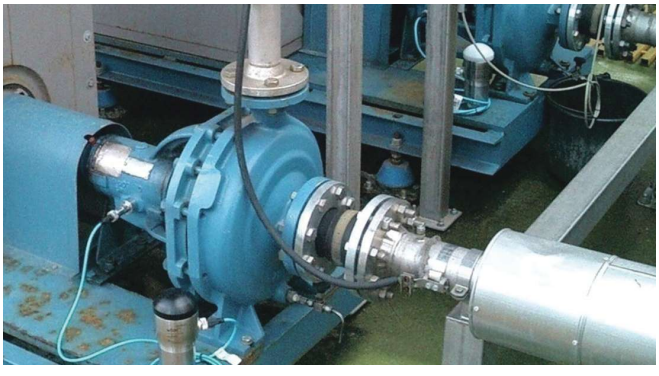


Temperature monitoring WSN powered by Micropelt's thermal harvester.



BE Bently Nevada vibration energy-harvester-powered wireless sensor node installed for machinery condition monitoring at Shell's Nyhamna Gas plant.

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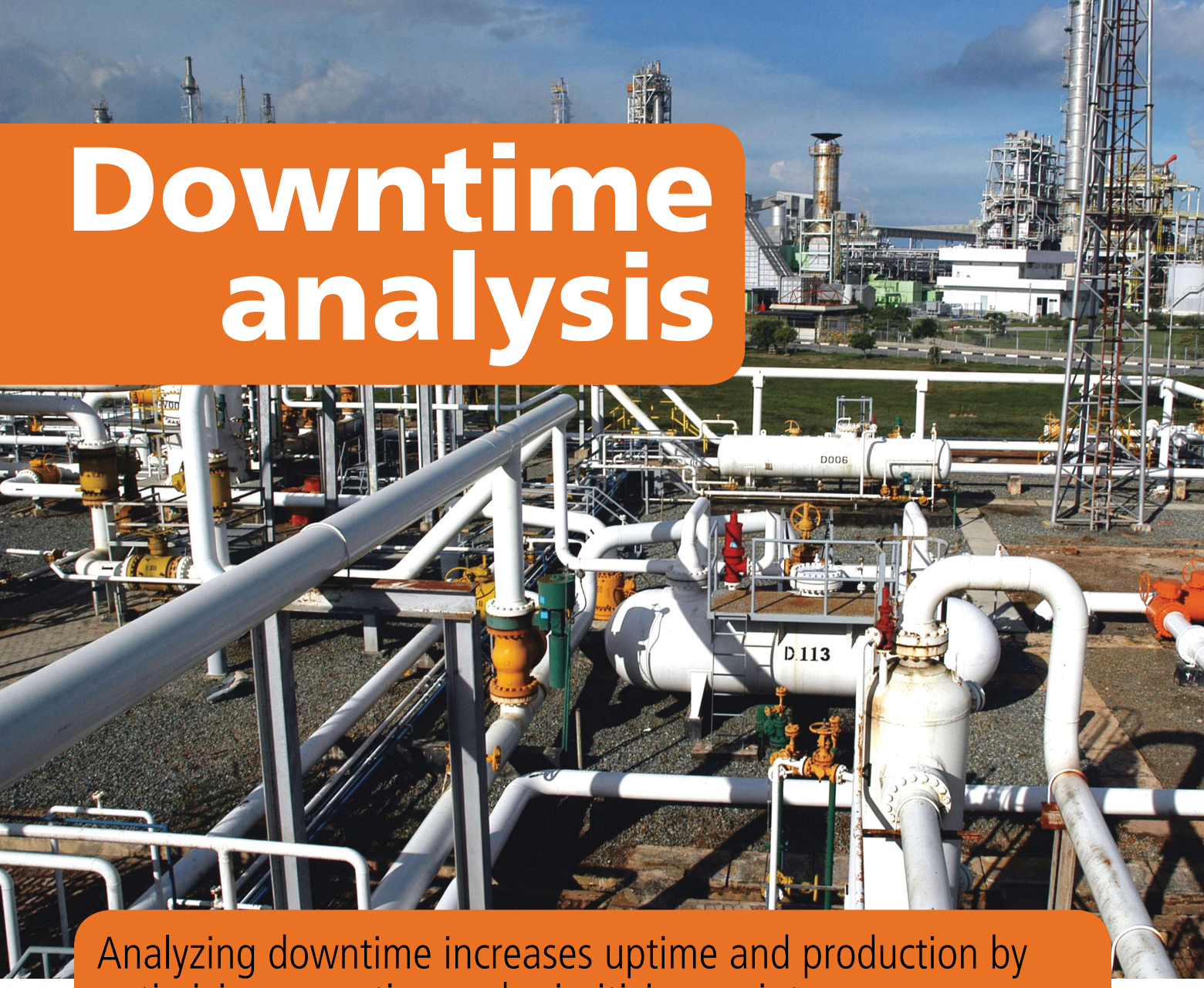
Downtime analysis



Analyzing downtime increases uptime and production by optimizing operation and prioritizing maintenance

By Wayne Matthews

Unplanned production stoppages and rate loss can have an enormous impact on the productivity and profitability of process plants. Rate loss is defined as the steady-state deviation in actual output from the rated maximum.

Downtime analysis (DTA) is an essential part of plant operations management, as it provides a powerful tool that enables a better understanding of the underlying issues that affect plant availability and rate loss. DTA enables identification and quantification of lost production capacity by accurately collecting data and measuring actual overall output against theoretical or rated capacity.

DTA systems provide easy-to-understand and comprehensive Key Performance Indicators (KPIs), which show the causes, duration, and timing of downtime and rate losses. Reports produced by DTA systems are used by production management and process engineers to implement continuous process improvement programs that can reduce the level of downtime and increase production rates.

Real-time reporting and analysis of KPIs enable the true impact of production interruptions to be determined and corrected—improving return on assets and availability, while increasing utilization of critical production processes. The costs attributable to unplanned downtime can be enormous, and for high throughput operations such as refineries, the cost of downtime

As with many process plants, downtime incidents at this fertilizer plant are very costly. Downtime analysis software and systems can reduce these incidents to increase uptime and throughput.

FAST FORWARD

- Downtime analysis is often performed manually in many process plants with limited results.
- Automated downtime analysis supplemented by manual identification of causes has proven to be a superior method for capturing downtime and rate loss information.
- Automated downtime analysis software presents results in easily understood terms that facilitate analysis of results and subsequent corrective action.

can run into the hundreds of thousands of dollars per hour.

DTA is typically provided via a vendor-supplied software program, but in addition can also include services in which process experts assist customers in interpreting the KPIs and then make recommendations as to best courses of action. DTA software is usually run on a PC, and typically communicates with the host automation system via an industry standard protocol such as OPC to gather the required data. This data is then analyzed and presented to plant personnel via printed reports and the automation system HMIs—or via viewing platforms such as office PCs, smart phones, and other web-enabled devices.

DTA provides a complete record of lost production by utilizing automated processes to capture all production stoppages and slowdowns. With some DTA software packages, operators can manually attribute each downtime or rate loss event to a specific cause or causes, adding valuable additional information, which complements the automatically gathered data. These records provide a holistic view of all relevant information, key to diagnosing underlying issues and providing effective solutions.

Typical functionality

A DTA system needs to have functionality that is appropriate for the particular type of industry in which it is used. For example, a downtime system intended for use in the process industries should include the ability to measure rate loss. These hidden losses are often overlooked or difficult to see, but can have a substantial impact on overall plant productivity.

DTA provides KPIs for lost production over a chosen time period and also the latest Downtime and Rate Loss events. This enables comparisons to be made between different production lines and shifts over a chosen time period. Lost production can be measured either as Downtime (on a time basis) or as Rate Loss (measured by production quantity lost). Downtime information can be translated into Rate Loss (and vice versa) based on assigned plant and unit capacities.

DTA enables staff to add reasons for each event, adding valuable understanding to the underlying reasons for lost production. Losses can also be attributed to multiple reasons, dividing the total loss by assigned percentages.

Downtime can still appear in reports even if a reason has not been entered. Downtime Losses can be automatically collected and registered on a per-plant basis, while Rate Loss can be calculated and registered on a daily basis per plant item. DTA results can be stored in the Plant Historian to ensure a common data set with other production performance information.

Typical DTA system reports include a summary report, reason classifications, downtime rankings, and rate loss rankings. Users can usually view reports by day, week, month, quarter, or year. Custom reports can often also be created within Microsoft Excel and/or through customer database queries. These and standard reports can generally be scheduled for e-mailing to users.

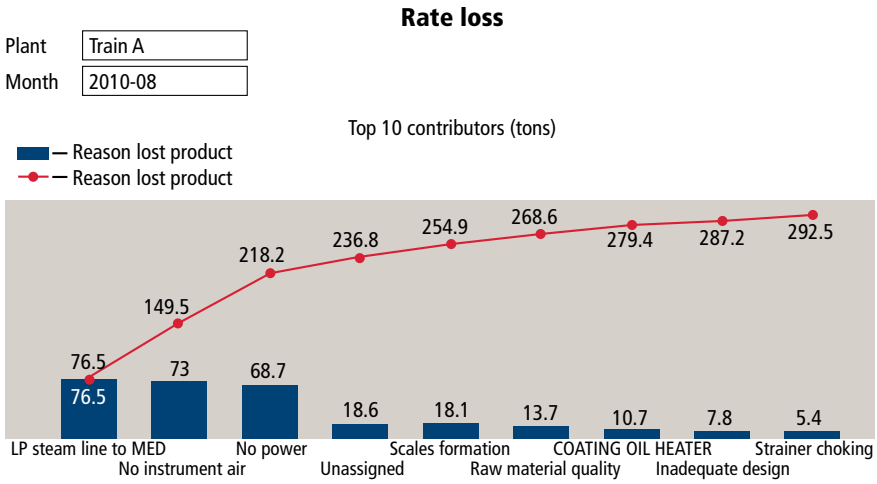
Why analyze?

Like any hardware or software product added to the basic regulatory control system, DTA requires expense and effort in implementation. But a correctly designed, implemented, and operated DTA system will provide a host of benefits as described below.

1. Maximizes return on assets

Maximizing asset utilization and plant productivity are keys to the profitability of process operations. Successfully identifying production underperformance and implementing process improvements forms the underlying basis for achieving these goals.

DTA helps achieve these objectives by identifying non-productive times when the plant



Reason	Contribution to loss
LP steam line to MED	76.5
No instrument air	73
No power	68.7
Unassigned	18.6
Scales formation	18.1
Raw material quality	13.7
COATING OIL HEATER	10.7
Inadequate design	7.8
Straining Choking	5.4

Page 1 of 1 Yokogawa Exaquantum/DTA

Rate loss, depicted on this screen shot, is defined as the steady-state deviation in actual output from the rated maximum. Without downtime analysis, these losses are difficult to detect.

is not running at its rated throughput. The latter issue can be easily missed with staff unaware of the level of lost production from running the plant at less than rated capacity.

2. Identifies common equipment failures

DTA causes can be grouped together to create a hierarchy of faults with major classifications such as mechanical, electrical, raw materials, procedural, personnel, and availability being expanded into more detailed causes.

As major causes of downtime are uncovered and underlying reasons addressed, the focus on particular downtime causes will change with some categories being expanded, with less significant causes being grouped together.

3. Reduces unplanned stoppages and production slowdowns

Using DTA, process engineers can gain a clear understanding of the most important causes of unplanned stoppages and production slowdowns. Often it is the short regular incidents that have the greatest cumulative effect on downtime. Large unusual events will certainly attract the attention of staff, but if these occur infrequently due to exceptional conditions, the cost of prevention may outweigh the benefit gained.

4. Compares existing and past performance

Comparisons can be made between current and past performance. Today's process plants are rarely run steady-state as there are often continual changes in raw materials, operating conditions, and end products. As a result, the impact of these changes needs to be understood and quantified. For example, a plant's steady-state throughput may be increased, but this may come at the cost of excessive downtime.

5. Generates accurate actionable information

The automated collection of downtime information, verified and enhanced with manual input, ensures all downtime and rate loss events are recorded without undue bias from operators or other staff. DTA provides quantifiable, accurate information that directs users to production areas and processes that need attention. Traditional manual recording techniques are highly prone to staff bias and incomplete recording of incidents.

6. Identifies areas for improvement

When a DTA is implemented, the most significant causes of downtime become readily apparent. DTA systems commonly provide a list of the top downtime causes and show the time, number of occurrences, and percentage of total time for each downtime cause. Using this information, underlying causes can be identified and a case developed for any necessary action, weighing the additional profit from increased production against the costs involved in rectifying the problem.

7. Prioritizes maintenance

By identifying the causes and effects of production loss, plant management can make informed decisions on maintenance priorities and schedules. In addition, when DTA systems are installed, the level of maintenance required typically decreases due to an improved understanding of the process and to better overall operation of the plant.

8. Improves operations and shared best production practices

The findings from DTA can help plant management make informed decisions on a range of practices including spares holding, operator training (e.g., enabling them to deal with incidents directly rather than calling in external help, leading to a quicker resolution of the problem), and operational procedures that can mitigate or eradicate causes of downtime. DTA findings also facilitate the sharing of best practices to reduce potential downtime.

9. Optimizes planning and scheduling

Regular unplanned stoppages and production slowdowns result in constant reshuffling of production plans and schedules. Reducing the level of these interruptions leads to more consistent production schedules and more accurate delivery forecasts.

10. Provides continuous plant availability improvement

DTA is an important tool for any continuous improvement program such as Six Sigma, as it provides a clear guide to where production is being lost, and where operational changes are needed to increase availability.

DTA benefits are substantial, but implementation is not always straightforward.

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Implementation challenges

Although DTA is a powerful tool with a wide body of proven applications, there are challenges to implementation:

1. Integration of existing automation systems with the DTA system

A successful DTA system relies on accurate data from the plant's regulatory control system. To correctly register a downtime or rate loss event, the DTA system needs a clear and unambiguous signal concerning the state of the equipment. These signals are typically calculated tags derived from raw tag data provided by the control system. While it may appear easy to determine whether or not a production unit has stopped, this is not always the case as manual overrides can mask true conditions.

2. Management and staff buy-in

Gaining Management and Operations staff buy-in is crucial to the success of a DTA project. At the management level, a product champion is needed to be the prime decision maker and move the project forward, ensuring the necessary resources and training are provided. This person will also drive the project forward by setting and evaluating objectives and system usage.

At the operator level, developing an operator-friendly system is equally crucial. Operators are already under significant pressure, so they must see any additional tasks as assisting them rather than adding to their burden. While events are captured automatically, it is the operators who manually attribute each event to one or more causes. If operators believe the downtime system is too onerous, for example in classifying downtime causes, they will be tempted not to attribute causes or to do so in a haphazard manner. This will result in a significant number of downtime events that are misattributed.

3. Downtime classification choices

DTA is typically implemented as part of an improvement project. As a result, operations management will know approximately where the problem lies. The challenge becomes designing a useful classification list and hierarchy of downtime or rate loss causes. The temptation is to create a very detailed listing covering every possibility. Doing so will cause difficulties in attributing causes from very long drop-down lists, and will not add useful information to results since most possible downtime events occur infrequently.

4. Quantifying the cost of downtime in lost production

In order to calculate the impact of downtime, it is necessary to understand the cost to the business of consequent lost production. Establishing this enables accurate ROIs to be calculated for proposed action plans.

5. Follow through to make sure action items are performed

A DTA system will help uncover the underlying causes of downtime and rate loss events. It is the responsibility of plant management, and the DTA champion in particular, to ensure this knowledge is translated into a plan of action that is carried through. Linking the DTA system to a continuous improvement program provides the motivation and accountability that an action plan requires.

A pilot project can be a good way to address challenges, as it allows the DTA system to be tested before it is rolled out across the entire plant. During the pilot project, inevitable implementation issues can be resolved, and experience can be gained in assigning the most appropriate downtime classifications. The pilot will also provide useful training and experience to the continuous improvement team prior to full-scale implementation.

From theory to practice

DTA was implemented for a major Middle East petrochemical producer on a world-scale facility producing a range of fertilizer products. The DTA package was developed to meet part of the requirement for a complete instrumentation and control system.

Due to the scale and complexity of the production facilities, restarting even a single production unit following a shutdown could take several shifts to complete. Bringing a plant back to full capacity would take days and possibly weeks, with significant problems associated with inevitable pipe and equipment blockages. In addition, while the plant was being restarted, it would be producing off-spec material, which would either need to be disposed of or reworked, with both of these activities being costly and wasteful.

Factors that influenced the decision to include DTA in the plant's automation system included the need to ensure product quality and consistency, a strong desire to quickly resolve the inevitable issues that attend the commissioning and start-up of a plant, and the need to ensure power and utilities maintained 100% continuity of supply.

When downtime or rate loss incidents occur, the DTA system captures these incidents and allows plant personnel to attribute each incident to specific causes. This knowledge is helping the plant to reduce the frequency of future events and incidents—while providing invaluable insights that will allow production, maintenance, and engineering teams to bring the plant to its full output potential.

DTA is an essential part of a continuous plant improvement program, as it provides a powerful tool that enables a better understanding of the underlying issues that affect plant availability and rate loss. The costs of unplanned downtime can be enormous, so even small reductions in downtime can have significant benefits to plant profitability.

Implementing a DTA system requires buy-in at management and operations level if it is to succeed, and incorporating DTA into a continuous improvement program helps ensure actions identified by the system are implemented.

ABOUT THE AUTHOR

Wayne Matthews is technical director for Yokogawa Marex Ltd., responsible for consultancy services, software development, systems integration, sales, and marketing. Matthews has been with Yokogawa for more than 18 years and in his present role for seven years. His background is in designing, implementing, and deploying manufacturing execution and plant information management software and systems.

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Martin Klein with the first commercial towed side-scan sonar, Boston Harbor, 1966.

Sonar advances, underwater discoveries earn Klein ISA Beckman Award

Martin Klein with students at the MATE-ROV Competition held at MIT in 2003.

By Jim Strothman

Fifty years ago, Martin Klein, then a young electrical engineering student at Massachusetts Institute of Technology (MIT), needed a subject for a thesis.

He "kind of stumbled" into the MIT lab of Dr. Harold E. "Doc" Edgerton, the "E" in EG&G International, Inc. "I asked if he had anything interesting to work on," Klein recalled.

"My life changed forever that day," he said.

Edgerton, who had been doing underwater photography with Woods Hole Oceanographic Institution since the late 1930s, had begun experimenting with sonar technology in the 1950s while working with famed oceanographer Jacques Cousteau.

Cousteau was taking deep-underwater photos in the Mediterranean. "In order to position that camera in the deep sea, (Edgerton) developed a device that produced two main signals that were tracked on a special recorder," Klein said. "Edgerton observed that the device showed the bottom, and also the geology under the bottom."

Mud penetrator launched career

"He started working on a gadget called the mud penetrator, and by a quirk of fate, I started assisting on this project." Klein significantly improved the device's signal clarity, delighting Edgerton.

In the years that followed, Klein developed—then continually improved—an instrumentation technology that forever changed underwater exploration.

Called commercial dual-channel side-scan sonar, the technology enabled ocean explorers to find the *Titanic* in 1985; *USS Monitor*; one of the most preserved War of 1812 ships sunk in Lake Ontario; and Benedict Arnold's gunboat in Lake Champlain, among many other shipwrecks.

Side-scan sonar also was used to find the remains of the Space Shuttle Challenger and downed aircraft, including TWA Flight 800, Swiss Air Flight 111, and John F. Kennedy Jr.'s plane off the Massachusetts coast.

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'Towfish' linked to surface recorder

Klein's technology typically consists of a torpedo-looking underwater device, dubbed a "towfish." Instrumented with transducers, the sonar device is attached via cable to a recorder aboard a surface ship.

"We've used the systems on big ships and small boats. I've used canoes, sailboats … all manner of vessels," Klein said.

Today, many nations' Navies use side-scan sonar, as do oil companies determining where to build pipelines. The sonar is also used in marine geology, hydrography, environmental studies, fisheries, dredging, and engineering projects.

Explored Loch Ness

In 1970, Klein teamed with Robert H. Rines, a famous patent attorney and MIT teacher of patent law. Rines had a fascination with Loch Ness' reputed monster. Also an inventor and holder of numerous technical patents, Rines in 1963 founded the Academy of Applied Science.

"We never did see a monster," Klein recalled, "but we saw large moving objects in the Loch." He acknowledges they "may have been tree branches or clumps of algae."

However, "we did find significant things," he said. His side-scan sonar showed large caves existed in the Loch's steep walls. In addition, "we found stone circle formations, which may have been made by an ancient civilization—maybe 100 feet in diameter—across the length and width of the Loch."

By accident, "we also found a twin-engine British Wellington Bomber Aircraft that went down in World War II," Klein said. It was raised and is now on display in an England museum.

Wins Beckman Award

Recognizing the significance of his technology, ISA will honor Klein by presenting him with its Arnold O. Beckman Founder Award on 17 October, the opening day of ISA Automation Week.

Given in honor of Dr. Arnold O. Beckman, founder of Beckman Instruments, the award "recognizes a significant technological contribution to the conception and implementation of a new principle of instrument design, development, or application."

Klein's citation credits him "for the invention and development of the dual-channel side-scan sonar instrumentation, which opened the world's oceans for exploration, safe navigation, and underwater recovery."

"I'm honored and humbled," said Klein, who left EG&G in 1967 to form his own company, Klein Associates, Inc. He started it in a basement of his rented apartment, and then later moved to a lumberyard he converted in Salem, N.H.—well aware his sonar-manufacturing competitors were giant defense firms with deep pockets, including EG&G and Westinghouse.

"Because of my field, I'm involved in many different worlds. In some, I am well known, in others, not at all. But like Beckman, I've always felt of myself as an instrument man," he said.

"We made a difference in opening up ocean exploration," Klein humbly said.

Award draws accolades

"Marty was the first to envision combining side-scan sonar technology with sub-bottom profiling, manufacturing what would be known as the Klein Tri-Fish," said Garry Kozak, who became a customer in 1974 and several years later became a Klein Associates employee.

Michael Fedenyszen, an I&C engineer with Vanderweil Engineers and ISA Boston Section Nomination Committee chair, nominated Klein for the Beckman award and obtained numerous letters from past and present Klein professional colleagues supporting Klein's nomination.

"Today, side-scan sonar instrumentation is used by the U.S. government, corporations, research institutions, and marine archaeologists around the world to map ocean floors, lakes, and river beds and to find objects of great interest and value," Fedenyszen said.

"Martin Klein was the unanimous choice of the (ISA Honors & Awards) subcommittee for the Beckman Founders Award," said Alan McMurry, subcommittee chair. "His contribution to the automation field with his side-scan sonar invention clearly meets the award criteria."

"Underwater archaeologists are indebted to Martin Klein and his instrumentation. There is no single piece of remote sensing equipment that has been used more by underwater archeologists than Klein side-scan sonar," said Joseph W. Zarzynski, a board-certified underwater archaeologist and executive-director of Bateaux Below, Inc., Wilton, N.Y.

Received many honors

Last March, the Boston Sea Rovers, a pioneer diving organization, named Klein "2011 Diver of the Year." This year also is MIT's 150th anniversary, and a special exhibition at the MIT Museum includes one of Klein's side-scan sonars.

In 2006, he received one of the Marine Technology Society/IEEE's highest honors, the Compass Distinguished Achievement Award.

An ISA Senior Life Member, he also is a LIFE Member of IEEE and the Navy League. He holds several U.S. and U.K. patents related to his technology and authored or co-authored numerous papers and articles for technical journals about the technology and underwater explorations.

Mentors young inventors

Mirroring his own mentor, "Doc" Edgerton, Klein since 2003 has volunteered his time to judge and mentor at regional and international remotely operated vehicle (ROV) competitions sponsored by the Marine Advanced Technology Education (MATE) Center, based in Monterey, Calif.

Funded in part by the National Science Foundation, the MATE competition challenges grade 5 through college students worldwide to work in teams developing ROVs.

Klein "inspired and encouraged (students), as well as teachers and parents who participated, to pursue their passions, seek the knowledge and skills they need, and remain committed," said Jill Zande, MATE Center associate director and competition coordinator.

"One of my goals in life is to help students and entrepreneurs starting their own businesses," Klein said.

"I come back from those meetings saying, 'the world is OK. (The students) are bright and have a positive attitude'. It's fun!"

"I believe (Klein's) legacy will be his technology developments as well as his giving back and encouraging future pioneers in the field," Zande said.

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Championing better education and training tools earns Wilbanks ISA Life Achievement Award

By Jim Strothman

Even before being promoted in 1973 to engineering manager – Control Systems at engineering and construction giant Rust Engineering Co., W. Gerald Wilbanks realized better education and training tools for young instrumentation engineers and technicians were sorely needed.

"In those days, most all instrumentation knowledge had to be developed by experience and self-studies," said Wilbanks, P.E., recalling his experiences after earning his electrical engineering degree at Mississippi State University in 1964 and beginning his career as an electrical controls engineer for Union Carbide Corp. in Port Lavaca, Tex.

"A lot of my early education came from ISA. I joined ISA in 1966, and when I moved to Rust Engineering in 1968 and got promoted to engineering manager – Control Systems five years later, we were hiring additional people, and I realized we needed to better train young graduate engineers," he said.

"So, we developed an educational lab to enhance measurement and other skills, developing lecture material and lab exercises. I got involved with the local ISA section teaching a three-day course on fundamentals."

Elected 50th ISA President

Wilbanks, soon after, became active with ISA at the national level, then international level. He became ISA's 50th president in 1995, the year ISA's Certified Control Systems Technician program was launched. For four decades, Wilbanks has worked at improving education and training tools and teaching courses. His efforts helped hundreds of engineers and technicians improve skills and advance their careers.

While ISA has been his main focus, Wilbanks has also been involved in examinations and other activities with the Alabama Society of Professional Engineers (ASPE), National Council of Examiners for Engineers and Surveyors (NCEES), and several other professional organizations.

Two years ago, Alabama Governor Bob Riley appointed Wilbanks to the Alabama State Board of Licensure for Professional Engineers and Land Surveyors. He currently serves as the board's secretary.

Recognizing his numerous accomplishments, ISA will honor Wilbanks by presenting him with the Society's prestigious Life Achievement Award on 17 October, the opening day of ISA Automation Week in Mobile, Ala.

Recognizes 'sustained dedication'

The award "recognizes individuals with a history of sustained dedication to the instrumentation, systems, and automation community."

Wilbanks' award citation reads: "In recognition of a lifetime of dedication to technical education, professional practice, certification and credentialing, and leadership to industry, the automation community at large, and his beloved vehicle for sharing the knowledge, mentoring, and leadership extension: ISA."

"The subcommittee unanimously selected Gerald Wilbanks because of the collective personal knowledge of Gerald's commitment to the automation profession and how he manifested his exemplary efforts to the benefit of ISA. In short, Gerald has always been a strong section mentor and supporter, leading the Birmingham (Ala.) Section along with a host of motivated local leaders over a long period of time to be one of the most successful in (ISA)," said Steve Huffman, chair of the ISA Honors & Awards subcommittee, which recommended Wilbanks receive the Life Achievement Award.

Asked his reaction when learning he won the award, "I was completely speechless and humbled," Wilbanks said. "I did not know my name was even submitted for that award."

Two years ago, ISA's Birmingham section honored Wilbanks by creating the "Daris and Gerald Wilbanks Endowment," which provides scholarships through the ISA Educational Foundation Scholarship fund. He gives much credit for his successful professional career to his wife, Daris, his bride of almost 51 years and to his son, Scott, and daughter, Lisa.

Developed CSE exam course

Nearly all U.S. states require individuals to pass a Control Systems Engineer (CSE) exam in order to become a licensed CSE professional. Around the year 2000, Wilbanks determined several local ISA chapters offered courses that trained engineers to pass pieces of the exam. However, ISA offered no structured, national-level course specifically designed to help engineers pass state exams and become licensed CSEs.

"You know how it is when you suggest something. You get to do it," Wilbanks laughed. "I spent a year, working with my (business) partner, sandwiching together pieces, extracting parts of ISA classes, developing a three-day ISA course. Titled "Control Systems Professional Engineer Exam Review," the course was launched in 2002. Since its inception, Wilbanks has taught the course several times a year in regional locations throughout the U.S. Also, he instructs a six-part web seminar on the same subject each summer as a part of the distance learning efforts of the ISA Training Institute.

"If I had to pick out a singular accomplishment of which I'm most proud, I would pick out that," he said. "I get a lot of letters from students thanking me for developing the class and teaching the class."

While ISA President and during his years on the Society's Executive Committee in the mid-1990s, he helped make ISA a more international organization. He served as a member of the China Instrument Society Liaison Committee, as chair and member of the Conference & Exhibit Global Oversight Board, and as chair and member of ISA's Globalization Development Council.

Consulting firm

Wilbanks has his own consulting firm, Documentation & Engineering Services, based in Trussville, Ala. As its principal engineer, he provides training, consulting services, auditing assistance, and design engineering to the industrial community.

"I always wanted to do two things: own my own business and be in sales and marketing. About 2004, my son and I acquired an existing manufacturers representative firm, so I accomplished my two goals. I had the excitement of meeting payrolls, doing sales, keeping up with paperwork and licenses," he said. After four years, "we sold (the firm) to a company in North Carolina. My son still works there, as a vice president, and I was able to make a dime or two and ride off into the setting sun."

Wilbanks worked for Rust Engineering for 32 years (from 1968 until 2000). During the 19 years he served as engineering manager – Control Systems, his department had nearly 200 employees working on a myriad of projects. Motivating him to improve training was the fact that "the better they were trained, the better job they did. And the better job they did, the better job the bosses would say I did."

ISA also honored Wilbanks in 2002, recognizing him as an ISA Fellow. In 1991, he was named Birmingham Alabama Society of Professional Engineers (ASPE) Engineer of the Year, and that same year, he was named Birmingham Area Engineer of the Year by the Engineering Council of Birmingham (ECOB). He served as ECOB's president in 1978.

"All of this (ISA, career, and other professional accomplishments) is tied to my passion for training," Wilbanks said. "I get an uplifting experience when I see young people who I helped train and develop move ahead and advance in their professional careers."

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Chatha, ARC Advisory Group Founder, wins highest ISA award—Honorary Member

By Jim Strothman

Twenty-five years after founding research and consulting firm, ARC Advisory Group, Andy Chatha is still thinking about industry's future.

When he and other ARC consultants predict what manufacturing technologies and business infrastructure concepts will be important near-term and well into the future, virtually all major industry suppliers and end users listen.

"We do a lot of thinking about the future," said ARC's founder and CEO. "Looking five years out is typical for our market forecasts, but we're also constantly thinking about and researching new concepts, new technologies, and new processes, and how companies can use new and existing enabling technologies to become more competitive. Increasingly, we've been looking at the human side of the equation, as well," Chatha said.

ARC was a one-man research operation when Chatha started the industrial consulting and forecasting business in 1986. Since then, Dedham, Mass.-based ARC has grown to nearly 80 employees with offices in North America, Europe, Japan, India, China, and Brazil.

Prestigious client list

ARC's clients include many Fortune 1000 user companies and other industrial giants. The list includes 3M, ABB, BASF, Dow Chemical, DuPont, Emerson, ExxonMobil, General Dynamics, IBM, Invensys, Microsoft, Mitsubishi Electric, Nestlé, Oracle, Procter & Gamble, Rockwell Automation, SAP, Schneider Electric, Siemens, Yokogawa, and others.

Recognizing his contributions to the automation industry, ISA has awarded Chatha its highest honor—Honorary Member. The award recognizes "special individuals for support of, and/or contributions to the advancement of the arts and sciences of instrumentation, systems, and automation." The society will present the honor to Chatha on 17 October, the opening day of ISA Automation Week 2011 in Mobile, Ala. ARC industry experts will also be participating in the conference.

"The ISA Honors and Awards committee is pleased to bestow upon Andy Chatha the prestigious designation of Honorary Member. He is recognized for his many accomplishments to the field of automation," said Gerald W. Cockrell, committee chair, an ISA past president, and Professor Emeritus at Indiana State University.

Industry experience cited

Chatha's citation credits him for his "over 35 years' experience in enterprise applications and automation and as a successful CEO of the respected ARC; executive advisor to some of the largest companies in the world; market analyst in the automation industry; project manager, software engineer, and designer of automation systems."

"Aside from establishing ARC as a respected market research and consulting business and hosting the very successful annual ARC World Industry Forum, Andy has also worked closely

with ISA over the past few years by including ARC forum programs in Automation Week and other activities," said Steve Huffman, chair of ISA's Honorary Member subcommittee, which recommended Chatha.

"He is a supporter of the ISA core competencies and our efforts to get more definition and recognition for our automation profession," Huffman added.

Before founding ARC, Chatha was an engineer and marketing manager at The Foxboro Company (now Invensys) for six years (1979–1985) during the early-distributed control system (DCS) days.

He served as a project manager at Westinghouse Electric Co. for five years (1974–1979) helping implement a blast furnace and a rolling mill automation project for two steel makers. From 1970–1973, he was a project engineer at General Electric Co. in the U.K., working on a rolling mill project.

Receiving the recognition "is a great honor," Chatha said.

Organized in teams

ARC has many teams, which focus on specialized areas such as automation, asset management, field devices, supply chain management, enterprise software, product lifecycle management, and energy optimization, among others.

Clients, who typically pay an annual fee, receive weekly and monthly research reports. In addition, "Clients can call us at any time if they have an issue," Chatha said. "We provide consulting services and also host forums and other high-value events that clients can and do attend."

ARC's annual World Industry Forum in Orlando, Fla., typically attracts top industry executives from around the world. The next such forum will be 6–9 February 2012 at the Renaissance Seaworld Hotel.

"Our success is because of our people," Chatha said. ARC's analysts are "experts from the industry" who have spent most of their professional careers working with industrial companies on multiple business issues and technologies," he said.

Chatha is particularly proud of ARC's global perspective. Not coincidentally, he was born in India, where he earned an electrical engineering degree at Panjab University. He obtained a Master of Science degree in systems engineering at Queen Mary University in London, where he also lived several years working for GE. He earned his MBA at Boston University's School of Management.

Asked about today's "hot button" issues, Chatha listed energy management, cyber security, new production management technologies, lifecycle management, cloud computing and mobility.

Collaborative Process Automation Systems, a well-received ARC model created to encourage suppliers to develop easier-to-use, easier-to-implement automation systems, is also a major ARC focus, Chatha said. Another is Collaborative Value Networks, which encourages end users to develop collaborative, integrated networks with their suppliers.

Schneider Electric, Telemetry & Remote SCADA Solutions, is a global supplier of remote automation solutions for SCADA systems in oil and gas, water and electrical utilities applications. Solution components include Accutech wireless instrumentation, SCADAPack controllers, Trio long-range data radios, and ClearSCADA enterprise software. All products are engineered to operate in harsh, unattended environments delivering higher productivity and efficiency while reducing operational costs across a wide area infrastructure.

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ISA Automation Week Booth 415

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ISA Automation Week Booth 307

'Everything should be as simple as possible, but no simpler'

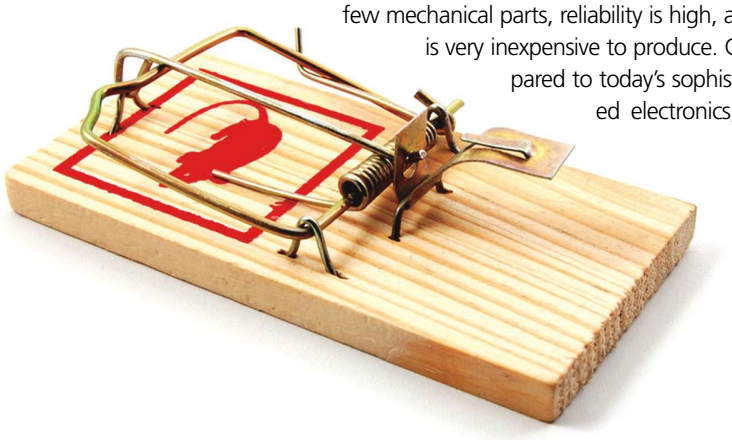
By Brian LaBelle

The abbreviated quote by Albert Einstein—"Everything should be as simple as possible, but no simpler"—can be used to illustrate an important contributor to business success.

In the industrial automation world, customers require products that fulfill a need or solve a problem. They want straightforward products, not over-engineered with unnecessary features that increase costs and reduce reliability. From the manufacturer's standpoint, providing products in the simplest form, which address that need or problem, is the first step toward success. How consistently the manufacturer accomplishes this is a strong determinant of success.

Customers consider several product aspects when buying—how simple the product is to use, how reliable it is, and how affordable it is. For the manufacturer, creating a simple product is not necessarily simple—it challenges the designer to be as efficient as possible, refraining from adding features that may be technologically impressive, but of peripheral value in meeting the need for which the product was intended. More often, extra features make the product more complicated to use, more unreliable, and more expensive. From a reliability standpoint, minimizing moving parts reduces the chance of malfunction. And a simple design will also result in the most cost-effective product to manufacture—fewer parts mean less materials and labor to produce.

The classic example illustrating the principle of product simplicity is the mousetrap. Invented in 1894, the mousetrap solves a precise problem by using a plain spring-loaded bar and trip mechanism. The simple design not only makes it very easy to use, it also results in two other important advantages. With few mechanical parts, reliability is high, and it is very inexpensive to produce. Compared to today's sophisticated electronics, the mousetrap appears elementary. The original design, however, has not changed for more than 100 years, and the product remains popular today. Its longevity proves that indeed sometimes less is more.



The product characteristics of simplicity, reliability, and affordability that affect the customer's buying decision can be extended far beyond the actual product, and when implemented effectively, can create the best customer experience possible. In terms of simplicity, how easy is it to do business with the manufacturer? Can the customer easily get the product through the distribution channel? How effective is the distributor in providing the product? And how easy is it to install and maintain the product? The answers to these questions determine how satisfying the customer experience will be.

Reliability is another product characteristic that can be extended to the total buying experience. A certain stock level in the distribution channel is essential to ensure quick delivery. How reliable is the distributor in stocking? Is the manufacturer's product documentation trustworthy? Do the manufacturer and distributor have a history of working together that the customer can rely upon?

Economy, another attribute derived from the simplicity concept, goes far beyond the initial product cost. How economical is the product to operate over time? Does it require frequent maintenance or service? If service is required, how expensive is it? Ultimately, the customer will determine which manufacturer best lives up to the promise of economical ownership.

In the end, it is not just the product that keeps the customer coming back, but the customer experience as a whole. To quote Einstein once again, "Any intelligent fool can make things bigger and more complex. … It takes a touch of genius to move in the opposite direction."

ABOUT THE AUTHOR

Brian LaBelle (Brian.LaBelle@georgfischer.com) is director of Marketing for GF Piping Systems (www.gfpiping.com), a provider of engineered solutions for the conveyance, measurement, and control of liquids. The company provides a range of piping products and instrumentation, including flow, pH, conductivity, pressure, temperature, level, turbidity, and chlorine monitors.

How much is downtime costing you?

By Dave Crumrine and Doug Post

Downtime costs every factory at least 5% of its productive capacity, and many lose up to 20%. But an estimated 80% of industrial facilities are unable to accurately estimate their total downtime cost (TDC). Many of these facilities are underestimating their downtime by 200-300% according to downtime consultants.

Not knowing your TDC compounds itself when you set priorities on capital investments. As your organization becomes more sophisticated at using financial tools, such as return on investment (ROI) and other leverage metrics, these tools become the key criteria in selecting and approving projects.

When ROI is used, it is especially important to know the real cost of downtime in your plant. By significantly underestimating it, you could be missing out on valuable opportunities for your own plant, making poor decisions, or neglecting what you intuitively know are the most important priorities. By knowing your TDC, you can pick the best capital projects and then make better decisions within these projects.

Sometimes the overall approach to a project can change based on this important number. It is not uncommon for the TDC on a retrofit project to approach or exceed the project's capital cost. In a situation like this, the right project delivery method and the right project delivery team are critical when executing an aggressive plan to minimize downtime. Selection decisions on your engineering, contracting, and other team support must be based on increasing your total project ROI (including reducing downtime and risk). This may be contrary to your normal purchasing methods. Keep your eye on the project ROI "ball" to overcome these hurdles to building a great team.

Many of the real costs of downtime are hidden in other cost areas and do not show up unless you account for them properly. To effectively calculate TDC, uncover all of these costs and list them in a separate "downtime" category.

Take a look at the important components of TDC. As you read the list, assess whether your downtime number fully includes these issues.

Equipment related costs

Annually calculated as a constant unit price

- **Labor cost:** Account for the full cost of direct and indirect labor with benefits, and include a share of all overhead positions in the plant, such as managers and support staff.
- **Product cost:** The cost per unit of production at each stage in the process, along with the units per hour at the machine/profit center, can tell you the value of the product lost during an incident.
- **Startup cost (per machine, line, cell, and profit center):** Include energy surge costs, setup (materials and manpower), percent of reduced production (units per hour lost), scrap produced (include recycle costs and/or scrap value), quality inspection and rework costs), as well as other startup costs.
- **Bottleneck cost:** Predict the cost impact on downstream equipment at each stage in the process.
- **Sales expectation:** Include the excess capacity, such as larger buildings, spare production equipment, etc.

Downtime costs

Per occurrence

- **Time:** Calculate and record the time from the first occurrence of equipment breakdown to the time when equipment was back in full production.
- **Reduced production**
- **Scrap**
- **Band-aid costs:** Figure in the costs of temporary fixes until the permanent fix is in place.
- **OEM, consulting, contractor costs:** Include the annual fee or estimated cost per year for support during downtime.
- **Tooling:** Calculate the replacement or rework cost for tooling (per occurrence).
- **Parts/Shipping cost**

As you can see, there are many factors to consider when determining TDC. With so much at stake, an accurate estimate of TDC is essential to your bottom line.

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Developing manufacturing skills for economic growth

By Emily Stover DeRocco

Over the past few months, manufacturing has enjoyed something of a national spotlight. It has been one of the few industries to show consistent growth, adding over 280,000 jobs in the past year and a half. And surveys show continued growth and confidence in the sector.

Manufacturing has been one of the few bright spots in an otherwise stagnant economy, and the newspapers and commentators have taken notice.

It is fitting manufacturing should now be getting such recognition because it is an industry that is truly vital to our economic security. No other industry creates more value or has a higher multiplier effect, and this results in a 17% compensation premium for manufacturing workers nationwide.

And it is possible more good news is on the horizon. Recent reports from two of the biggest consulting firms in the world, Boston Consulting Group and Accenture, looked at what is euphemistically being called *on-shoring*. What their research shows is manufacturers are discovering China is not as cheap as everyone thought. When you factor in everything from the shipping of goods to the availability of workers to the inflexibility of the supply chain and the manufacturing specifics, the cost of producing goods in the U.S. is actually very competitive with the Chinese cost.

However, human capital is one of the critical issues facing U.S. manufacturing. Between the coming renaissance in manufacturing and the impending baby-boomer retirement, manufacturers are going to have to fill millions of positions in the next decade.

In fact, we are already seeing the beginning of the problem today. In a recent nationwide survey by The Manufacturing Institute and Deloitte, 32% of manufacturers reported moderate to severe skills shortages—and this was in the summer of 2009, at the height of the recession and job losses.

The time is right for manufacturers to change the way they approach and manage their human capital. The Manufacturing Institute is rolling out the flagship education initiative of the manufacturing industry as the national solution to the talent challenge.

Between the coming renaissance in manufacturing and the impending baby-boomer retirement, manufacturers are going to have to fill millions of positions in the next decade.

In manufacturing, the core premise of this solution is there are standards for every imaginable input and output. Whether it is the composition of steel, the tolerance of machines, or the failure rate of a part, manufacturers can give the details to three decimal points. A manufacturing education and training system, then, should allow manufacturers to be as rigorous in the standards they apply to their most important asset—human capital.

These standards are not in the form used by traditional education, which measures seat time through credit hours. Instead, these standards are competency based, demonstrated through mastery, and verified through certification.

To develop the manufacturing talent solution, called the NAM-Endorsed Manufacturing Skills Certification System, The Manufacturing Institute joined with several other leading industry groups to create a system of nationally portable, industry-recognized credentials. These credentials and the training required to obtain them certify an individual possesses the basic skills required to work in any sector of the manufacturing industry.

The manufacturing system can be envisioned as a pyramid of skills certifications, with an initial focus on the skills required for all entry-level jobs in manufacturing today:

- Personal effectiveness skills
- Foundational academic competencies—for manufacturers, those are applied math, reading, locating, and using information
- General workplace competencies, which cover the fundamentals of business
- Industry-wide technical skills related to basic manufacturing processes, including production, logistics, machining, quality assurance, safety and health, and technology

The foundational competencies in the first tiers are grounded in ACT's National Career Readiness Certificate. The workplace and technical competencies are covered by the Manufacturing Skill Standards Council's Certified Production Technician, the National Institute for Metalworking Skills' Machining and Metalforming certifications, and the American Welding Society's Certified Welder series. Finally, the Society of Manufacturing En-

gineers' Engineering Technologist certification caps the entry-level skills system, recognizing the infusion of technology into all manufacturing processes.

The Institute also is developing higher-level pathways for sector-specific skills and competencies, including automation. We have recently announced a partnership with ISA to bring automation and control systems certifications into the Manufacturing Skills Certification System, adding ISA's Certified Control Systems Technician (CCST) and Certified Automation Professional (CAP) certification programs to the system's offerings.

The Skills Certification System is currently being implemented in community colleges' for-credit programs of study with connections down to high schools and technical schools and up into universities. The specific learning content needed to obtain the skills required to achieve each certification has been mapped to educational pathways. And these educational pathways are aligned to career pathways in quality jobs in manufacturing.

This system is not just a training program for manufacturers. It is the framework for building a workforce proficient in applied science, technology, engineering, and math (STEM) because as our economy continues to advance, more and more industries are going to require a STEM-capable workforce.

On 8 June, we stood with President Barack Obama and announced the goal of awarding 500,000 credentials for high-quality manufacturing jobs in the next five years. However, we cannot reach this goal without help from our nation's manufacturers. As we make significant progress aligning education with the needs of industry, we need industry to reflect this paradigm shift in their hiring practices. Manufacturers can accelerate these efforts, becoming strong advocates for building credentialed talent.

ABOUT THE AUTHOR

Emily Stover DeRocco is president of The Manufacturing Institute. This article was adapted from a speech given at the Lehigh Valley Manufacturing Summit in Allentown, Pa.

Thermocouples: What one needs to know

By Thomas W. Kerlin and Mitchell P. Johnson

Typical thermocouples are very simple, consisting only of two dissimilar conductors joined at one end and connected to instrumentation at the other end. Thermocouples are rugged and inexpensive. They are widely used and will continue to be. They typically provide satisfactory temperature measurements, but are not foolproof. Problems in applications can happen when users are unaware of some simple facts about thermocouple properties and principles or are careless or uninformed during installation.

Here are the 12 essential facts:

1. Thermocouple measurements have significant uncertainties due to manufacturing tolerances. For example, the tolerance on a standard grade Type K thermocouple at 1000°C (1832°F) is ±7.5°C (±13.5°F), indicating two sensors could differ by 15°C (27°F). Errors at the extreme of the tolerance range are possible but unlikely because manufacturers strive to build sensors with nominal calibrations.

2. Thermocouples do not produce a voltage at the junction. Rather, the voltage produced occurs along the length of wires that are in a temperature gradient.

3. Thermocouple thermometry requires measurement of the voltage produced by the thermocouple while no current flows in the circuit. (We need to have the "open circuit voltage.") Consequently, read-out instrumentation must have a large input impedance to adequately approximate open circuit conditions.

4. Thermocouples can decalibrate in use. This is usually a gradual process and can easily go unnoticed. Decalibration can impact process performance.

5. The most likely cause of decalibration is creation of inhomogeneous sections in one or more wires caused by chemical attack that alters the wire composition or mechanical effects that alter the wire metallurgy. Such an inhomogeneous section causes errors if, and only if, it experiences a temperature gradient.

6. Recalibration of used thermocouples is ineffective and a waste of time. Errors due to inhomogeneities, the likely cause of decalibration, depend on the temperature gradient when in use and duplicating that gradient in a calibration facility is not possible.

7. Metal-sheathed thermocouples and thermowells that house sensors conduct heat along their length. This can cause the sensor to read a temperature that lies between the process temperature and the temperature at the back end of the sensor. This problem increases with shorter, fatter sensors.

8. Sensors and thermowells can suffer mechanical failure due to vibration, stress, or pressure. Software is available to enable selection of components that are unlikely to experience these problems.

9. The time response of a sensor immersed in a process depends strongly on fluid conditions around the sensor. Time constant values reported by sensor manufacturers apply only for the conditions at which they made a measurement.

10. Thermocouple voltage depends on the temperature difference between the junction and the back end where the voltage is measured. But the thermocouple tables are based on a back end temperature of 0°C. Determining temperature requires compensation for departure of the back end temperature from 0°C. Dedicated thermocouple instruments handle this automatically, but using a voltmeter requires an understanding of the compensation procedure.

11. Thermocouple loop analysis is a simple method whose use explains all aspects of thermocouple use and misuse. Every serious user of thermocouples should learn and use this method.

12. The so-called "Laws of Thermoelectricity" should be forgotten. They have been part of the folklore for decades but are of little or no value in making good measurements. Thermocouple loop analysis is the way to go.

Thermocouple loop analysis

A homogeneous section of a conductor that experiences a temperature T_0 at one end and a temperature T_1 at the other end experiences a voltage difference, V, between the two ends. The voltage, V, is given by the following equation:

$$V = S\,(T_1 - T_0) \qquad (1)$$

where

S = the Seebeck coefficient (μV/°C)

The Seebeck coefficient (also called the "thermoelectric power") is the fundamental thermoelectric property related to thermocouple thermometry. It is a physical property of a material, like its density, thermal conductivity, or electrical resistivity. It is independent of the size and shape of the conductor but does vary with temperature. Because of this temperature dependence, the relation shown in Equation 1 is an approximation. This approximation is adequate for the qualitative analysis of thermocouple circuits but is inadequate for predicting the voltage that would be observed for a specific thermocouple in a specific temperature gradient. However, for understanding how various thermocouple configurations work, it is quite satisfactory.

The simple relation between voltage and temperature difference along the conductor may be used to predict thermocouple performance, analyze thermocouple configurations, and troubleshoot problems with thermocouple thermometry. This procedure is called thermocouple loop analysis. The procedure may be illustrated for the basic thermocouple shown in Figure 1. The approach is simply to sum up the voltage contributions for each homogeneous portion of the conductor. For example, if we choose to start the summing process at the open end of conductor A, the voltage is as follows:

$$V = S_A(T_1 - T_0) + S_B(T_0 - T_1) \qquad (2)$$

This is algebraically the same as

$$V = (S_A - S_B)(T_1 - T_0) \qquad (3)$$

Note the difference in the Seebeck coefficients for the two conductors appears in Equation 3. This always happens in thermocouple loop analysis, and it is

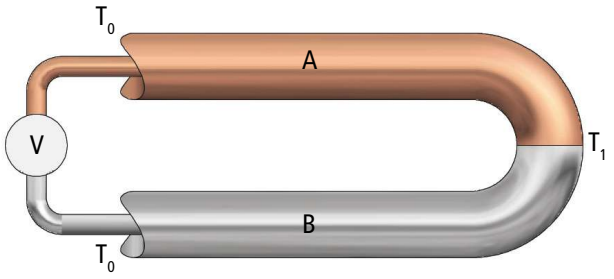

Figure 1

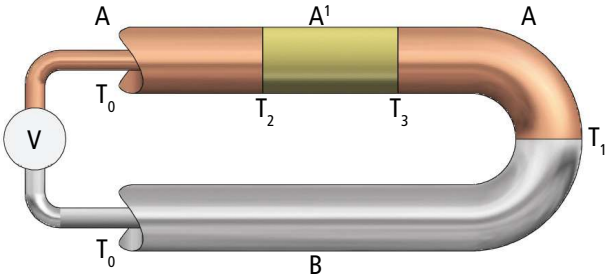

Figure 2

the property that is of practical interest in thermocouple thermometry. It is called the relative Seebeck coefficient (between material A and material B) and is written "S_{AB}." That is,

$$S_{AB} = S_A - S_B \quad (4)$$

Consequently, Equation 4 may be written as follows:

$$V = S_{AB}(T_1 - T_0) \quad (5)$$

This is the fundamental relation in thermocouple thermometry.

Thermocouple loop analysis enables the thermocouple user to characterize any thermocouple configuration. It explains the consequences of damage to any part of a thermocouple circuit.

Reference temperature compensation

The thermocouple tables and mathematical functions for voltage vs. temperature give temperature for measured voltage when the reference end (the point where the voltage is measured) is at 0°C. Since the reference temperature is not 0°C in typical applications, a correction must be made before determining the junction temperature. Using thermocouple loop analysis, we may write

$$V(T_2\text{-}0) = V(T_1\text{-}0) + V(T_2 - T_1). \quad (7)$$

That is, we must add the voltage that would have been observed if the junction was at temperature T_1 and the reference temperature was at 0°C (the first term in Equation 7). Thermocouple readout instruments perform this correction automatically. Of course the reference temperature, T_1, must be known. Instruments include a sensor (usually an integrated circuit sensor or a thermistor) to provide T_1. The instrument then determines the voltage, $V(T_1\text{-}0)$ from a stored formula for voltage as a function of temperature, and adds it to the measured voltage in order to obtain the voltage that would have been measured if the reference temperature was 0°C.

The inhomogeneity problem

In the case in which chemical or metallurgical changes occur only along a portion of the thermocouple wire, the Seebeck coefficient is unchanged, except over the length of wire where the chemical or metallurgical changes occurred. A simplified depiction of the situation is shown in Figure 2, where the changes in Seebeck coefficient occur abruptly. Thermocouple loop analysis gives the following:

$$V = S_{AB}(T_1 - T_0) + S_{A'A}(T_2 - T_3) \quad (7)$$

The first term is the voltage that would have been produced if the thermocouple had not undergone attack. Consequently, the second term is the error caused by the inhomogeneous region. If the relative Seebeck coefficient between the unaffected wire and the altered wire is nonzero, then a measurement error will occur if T_2 is not equal to T_3. That is:

An inhomogeneous section in a thermocouple wire will cause a measurement error if, and only if, it resides in a temperature gradient.

This is a very important result. Process or environmental conditions usually cause alterations only along some portion of a wire. This makes the measurement error dependent on the temperature profile along the wires. One consequence of this is it confounds any attempts to recalibrate used thermocouples. This is because in a calibration facility it is impossible to duplicate the temperature profile that the thermocouple system will experience when it is being used in a process.

The error caused by the development of an inhomogeneous section in a thermocouple circuit may be positive or negative. As can be seen in Equation 7, the polarity depends on the relative Seebeck coefficient between the affected and unaffected segments (since $S_{AA'} = S_A\text{-}S_{A'}$, $S_{AA'}$ can be positive or negative). Also, the polarity depends on the temperature difference, $T_2\text{-}T_3$, across the affected region, and this can be positive or negative.

Conclusions

Thermocouples are widely used and are here to stay. Problems are infrequent, but potentially serious. Users need to know how to use thermocouples properly and to troubleshoot effectively when problems arise. Thermocouple loop analysis is simple (eighth grade mathematics), comprehensive, and effective. It is an essential tool for all who are responsible for ensuring temperature measurements are correct.

ABOUT THE AUTHORS

Thomas W. Kerlin is Professor Emeritus from The College of Engineering at The University of Tennessee where he served as Professor and Head of The Nuclear Engineering Department before retirement. He has published numerous articles and two books on temperature measurement. **Mitchell P. Johnson** is president of JMS Southeast, Inc., a 31 year manufacturer of thermocouples and related products. He serves as a member of ISA, the ASTM Temperature Measurement committee, the ASME Thermowells committee and has published articles and presentations on temperature measurement through ISA.

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Proactive versus reactive standards for nuclear plant design

By Jeremy Shook and Mark Burzynski

Today, the U.S. nuclear industry is in the process of designing and licensing a new generation of nuclear power plants with modern, digital instrumentation and control (I&C) safety systems. These systems are being licensed against standards historically based on analog technology. At the same time, other process industries have adopted a newer set of standards for the design of safety instrumented systems (SIS) that reflect advances in I&C technology. While the nuclear industry has struggled in this area, the general process industry has been able to develop and implement standards for safety I&C systems that use various types of digital technology.

The ISA-84 series of standards addresses functional safety of safety instrumented systems through a more proactive approach, as opposed to the U.S. Nuclear Regulatory Commission's (NRC) reactive approach in its regulations. The NRC regulations regarding quality are high level in nature. However, there are other NRC requirements (10 CFR 50 Appendix B) and guidance documents (various regulatory guides, endorsed IEEE standards, and branch technical positions, or BTPs) on software quality management that would apply to a SIS. Take a look at how the two compare and contrast to get a broader perspective of both standards.

ANSI/ISA-84.00.01-2004 Part 1 (IEC 61511-1 MOD), Functional Safety: Safety Instrumented Systems for the Process Industry Sector - Part 1, is the primary standard in the U.S. for the design of SIS, which is based on IEC 61511. This standard is written specifically to deal with the types of technology typically found in I&C systems today, including electrical, electronic, and programmable electronic technology.

The scope includes requirements for the entire life cycle of a SIS, ensuring it can be confidently entrusted to place or maintain the process in a safe state. This starts with overall management of functional safety, going through SIS installation, operation, and decommissioning.

In comparison, the approach in the NRC requirements and guidance is generally a reactive approach, analyzing what is there to see if it is acceptable versus the ISA-84 approach of defining defense-in-depth diversity (D3) at the top and flowing it down.

The NRC's BTP 7-14 only addresses safety software and is generally silent on the broader life cycle in which the software must reside. By not addressing the whole system (while still recognizing the role that software plays as ISA-84 does), the remainder of the quality assurance under the NRC rules falls under 10 CFR 50 Appendix B, which is not well coordinated with BTP 7-14.

In contrast, ISA-84 considers the whole system, including sensors, logic solvers, and final elements. In addition, it provides a quality management and safety life-cycle framework for the entire system, while still recognizing the special role software plays. This approach is more integrated with greater assurance the SIS will perform its intended high-quality functions.

NRC requirements regarding D3 are also general in nature. General Design Criteria (GDC) 22 discusses the use of functional diversity to reduce the probability of a complete loss of the protection function. NRC guidance in BTP 7-19 provides more specifics with regard to acceptance criteria for software common cause failures.

ISA-84 deals with D3 proactively in two parts. First, the layers of protection (LOPs) are defined. The safety instrumented functions (SIFs) are defined as a result of the process hazard and risk assessment and are allocated to the various LOPs (i.e., defensive levels using nuclear terminology). In addition, the SIL level associated with each SIF is defined and allocated at the same time. This integrates the activities of the deterministic safety analysis and the probabilistic safety analyses performed in nuclear power plant design. Second, independence of the LOPs is specified and analyzed.

Reliability

NRC requirements regarding reliability include GDC 21 and IEEE Std 603 clauses 4i, 5.1, 5.6.1, and 5.15. While GDC 21 states "the protection system shall be designed for high functional reliability and in-service testability commensurate with the safety functions to be performed," in general, the reliability requirements are qualitative in nature and do not require quantitative reliability goals. Specifically, NRC requirements use the single failure criteria, along with independence between redundant divisions, to provide a certain measure of reliability in the performance of safety functions.

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ISA-84 uses a qualitative and a quantitative approach for reliability, using the SIL concept. In this concept, a specific quantitative reliability goal is established for each SIF based on the level of risk reduction required. In addition, clause 11.4 specifies minimum hardware tolerance for a given SIL. It is interesting to note while in some cases SIL 1 functions do not require any tolerance to hardware failures, SIL 4 functions (the level at which many reactor trips and engineered safety features actuations may be specified), would require tolerance to three hardware failures at a minimum.

The primary advantage of the ISA-84 approach is in the treatment of independence. In the NRC approach, reliability is provided by the single failure criteria, which is ensured by redundancy and independence. The challenge in many recent reviews of digital safety I&C designs is how to achieve and demonstrate independence. With the ISA-84 approach, reliability is defined quantitatively with the SIL criteria, along with a qualitative hardware fault tolerance. As such, independence is not a specific requirement, but the degree of independence between redundant elements either simplifies or complicates the reliability analysis, which is the true design objective.

Only two main areas are not specifically discussed in the ISA-84 framework—support system operation (such as electrical, cooling water, and HVAC) and post-accident operation. Both stem from the fundamental difference in nuclear plants from typical process facilities, which is the generation of decay heat after reactor shutdown, and thus the need for continued operation of the SIS after initial accident mitigation.

ABOUT THE AUTHORS

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This article was edited from a paper entitled, "An Evaluation of ISA84 for Use in the Design and Licensing of Nuclear Power Plants," presented at the 54th ISA POWID Symposium in June 2011. To read the entire paper, visit www.isa.org/link/Standards_INT.

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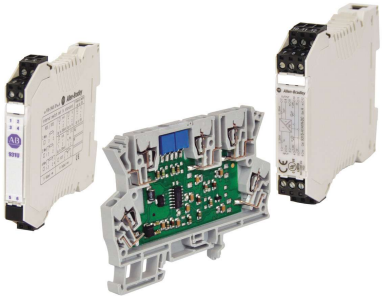
Signal conditioners



Comprised of 29 distinct products, the Allen-Bradley Bulletin 931 analog signal conditioners are designed for process applications and they isolate multiple signals on the same power source, reducing ground loop and common mode noise. By incorporating the Bulletin 931 analog signal conditioners into an existing control system, users can convert a wide range of signals from field devices into a standard 4 to 20 milliamp signal. The signal conditioners can be integrated with the PlantPAx Process Automation System, leveraging a single control platform for discrete, batch, process, safety, drives and motion control. The signal conditioners convert thermocouple signals into the standard 4 to 20 milliamp signal, which can be run over long distances on standard cable with less chance of signal deterioration.

Rockwell Automation, www.rockwellautomation.com/go/prsignal

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See us at ISA Automation Week—Booth 716

L-Series midrange controller

The L-Series is Mitsubishi's newest midrange controller that is designed specifically to fill the gap between many brick style microcontrollers and large modular rack-based systems. The L-Series is modular in design and can be easily expanded beyond its built-in I/O using a wide range of compact I/O modules and specialty modules that include analog, motion, and networking interfaces. The L-Series offers many other embedded features making it ideal for machine builders and most midrange applications.

Mitsubishi Electric, www.meau.com

See us at ISA Automation Week—Booth 415

PLC with Ethernet Remote I/O

The Modicon Quantum PLC now supports Ethernet Remote I/O (RIO). The 140CRP31200 and 140CRA31200 Ethernet head and drop modules allow customers to move to an open standard EtherNet/IP network. The architecture now is easier to implement, support and maintain. Larger amounts of data can be written to/read from the RIO drops reducing the number of I/O drops required and allowing higher density of analog modules in the remote racks.

Schneider Electric
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ARC Advisory Group helps clients get value from technology investments. ARC is the leading research and advisory firm for industry and infrastructure. For the complex business issues facing organizations today, our market and technology analysts have the industry knowledge and first-hand experience to help our clients find the best answers and avoid the mistakes others have made. Our comprehensive market reports provide accurate business intelligence for fact-based decision making.

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Designed with amperometric sensing technology, the Signet 4630 Free-Chlorine Analyzer System incorporates a clear flow cell, flow regulator, sensors, filter, and rotameter. The fully integrated system provides a turnkey solution for accurately measuring free chlorine. Applications include primary/secondary disinfection, water distribution, de-chlorination, algae growth prevention, slime (bio-film) control, and taste and odor control. The reagent-less system includes pre-wired electronics, a 120 VAC power plug, two 4 to 20 mA outputs and two mechanical relays.

GF Piping Systems
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Infrared temperature sensor

The infrared sensor with wireless transmitter features remote IR sensor and radio wireless transmitter in a NEMA enclosure. Each unit transmits process temperature, ambient temperature, signal strength, and battery status. This unit features an adjustable emissivity from .10 to 1.0, and one receiver has the ability to work with up to 48 transmitters. The low power operation and sleep mode feature allows for a long battery life. Package comes with free software that converts your PC into a multi-channel chart recorder or data logger. It interfaces with other Omega products, the UWTC-REC1 for multi-channel PC chart recording and data logging or UWTC-REC2 (Single Channel Industrial Transceiver with Analog Output and Alarm.)

OMEGA Engineering, www.omega.com

Polymeric tubing

The company has extruded tight tolerance PEEK tubing for years, and has recently added fluoropolymers to its product line. Included in this offering are FEP, PFA, and ETFE (Tefzel), in 360um, 1/32", 1/16" and 1/8" OD, with IDs from .004" to .095", dependent upon OD. Each is available in natural or a variety of colors. Custom sizes and colors are available in OEM quantities. PEEK tubing is available in a wide variety of forms including natural, solid color coded, dual layer color-coded, striped, and dash-stripe coded. The dual-layer color coded PEEK is recommended where maximum chemical resistance and biocompatibility are required. Tolerance is ±.0005" for IDs up to .010", and ±.001" for larger IDs.

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Industry PCs

The Industry PCs support the new IEEE 802.11n WLAN standard. The new standard features faster data transfer rates, better reach, more efficient energy consumption, and increased overall security for transmissions. Companies that have, or are planning for, wireless network connections for their industrial PCs will benefit. Compared to previous versions, the new IEEE 802.11n WLAN standard (also referred to as the n-standard) features significant overall improvements in data transfer processes. This n-standard is now supported by the company's industrial PCs. The greatest advantage is higher data transfer rates reaching gross bit rates of up to 450 MBit/s and a net bit rate of roughly 180 MBit/s. To optimize speed, users can employ either the 2.4-GHz or 5-GHz channel. To maintain uninterrupted transmission, the Industry PCs have up to three antennas that will receive and send information simultaneously.

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Emerson Process Management
www.emersonprocess.com

Circuit board terminals

Pluggable printed circuit board (PCB) terminals offer a 3.5 millimeter pitch high density PCB connector that provides an extremely high density design while still maintaining a pitch that is easy to work with. The angled PCB headers are particularly designed for contacting automation technology control devices or controllers. The newly designed PCB headers offer a particularly high contact density inside a compact housing. The two-tier and three-tier headers are available in two to 16 pole designs per tier, in straight or angled versions, with or without latching flanges for mating connectors. The three-tier version, wiecon 8513 SDGN, features a maximum of 48 poles for the connection of 1.5 mm² wires.

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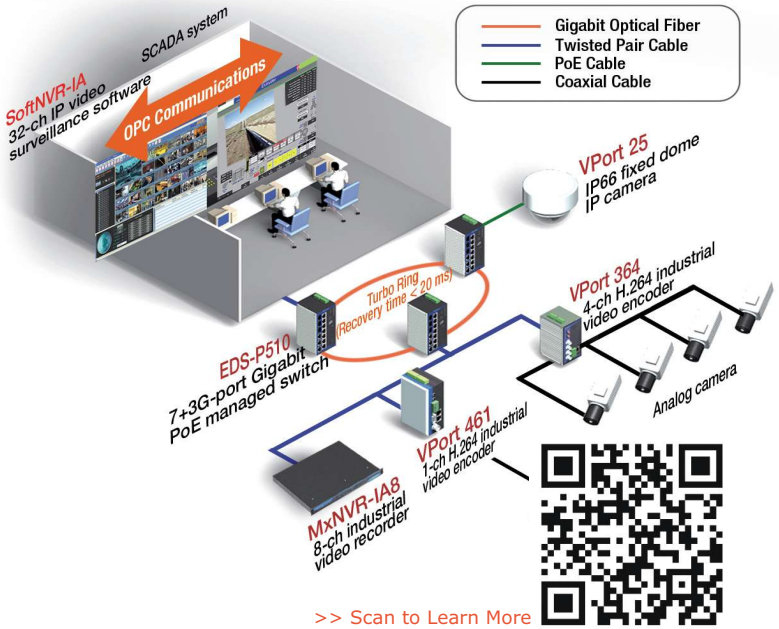

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Pathway to CAP and professional development

By Abdul Rauf

Editorial note: *This is the second of a three-part series on the Certified Automation Professional (CAP) program.*

With the growing production demands and need for standardization, automation system vendors are introducing more robust and efficient platforms to meet customer requirements. As they become more aware of their clients requirements, vendors also recognize the need for a consistent skill set for their employees. Since I joined the automation field, I have kept up with these growing technologies using any continuous education opportunities available. This led me to discover the Certified Automation Professional (CAP) program from ISA.

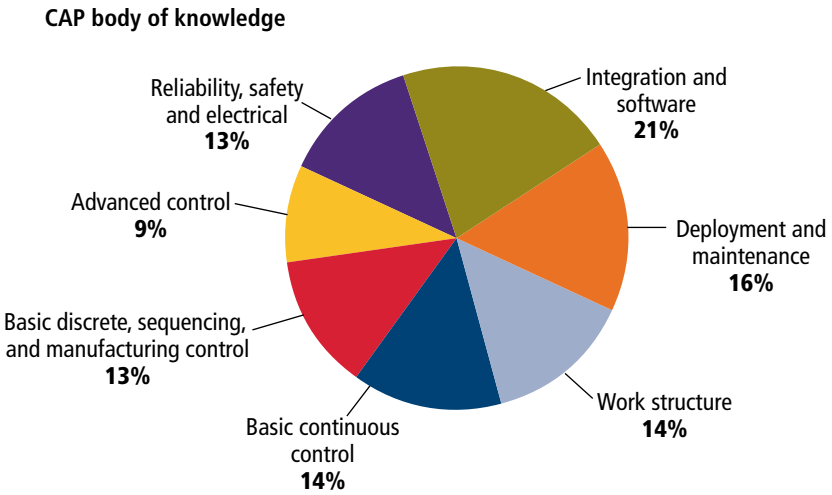
CAP helps automation professionals develop a strong knowledge base of a wide range of topics and the best standard practices used in industry. It was a perfect match to my career development needs and demands, so I decided to move ahead.

Eligibility

The first step to proceed in the CAP program is to check eligibility for certification. An applicant may fall in one of two categories:

1. If you hold a four-year technical degree, you will need five years of work experience or a total of 7,500 man hours in the field of automation.
2. If you hold a two-year degree or do not have a degree, you will need 10 years of work experience or a total of 15,000 man hours in the field of automation.

I had already completed 7,500 man hours at work, so I collected documentation of past projects and shared a summary with our Human Resource (HR) department. With consultation from my engineering manager, HR drafted a reference letter that verified my work experience, which I presented with my CAP application to ISA.



Distribution of topics covered in the CAP body of knowledge

If you do not have sufficient work experience and are a new graduate as well, you can still go for the CAP Associate program. The applicant has a six-month window of eligibility to take the exam, three months before or after the applicant's graduation date from the four-year degree program. A CAP Associate counts as 1,500 man hours of experience.

Application processing

Filling out the application for CAP is quite straightforward and hassle free. The online application at ISA's website allowed me to fill in personal details and professional work experience history. ISA uses this information to evaluate the candidate's eligibility for CAP. As a rule of thumb, one is not required to present proof of their work experience and academic credentials. The ISA randomly selects a number of applications and subjects them to an audit. In case an application is selected, proof of your credentials will need to be shared with ISA.

The application fee for CAP is $345. As an ISA member, I took advantage of their reduction on the application fee by $50. Payments can be made online using a credit card or via check/money order. For any related queries, you can call ISA at 919-549-8411.

Testing windows

Three testing windows have been made available throughout the year—each window lasting for about 60 days. The applicant is required to register before the exam application postmark deadline date to avail the respective testing window. I registered last year for 2010's Window 3: 1 November – December 2010.

Preparation

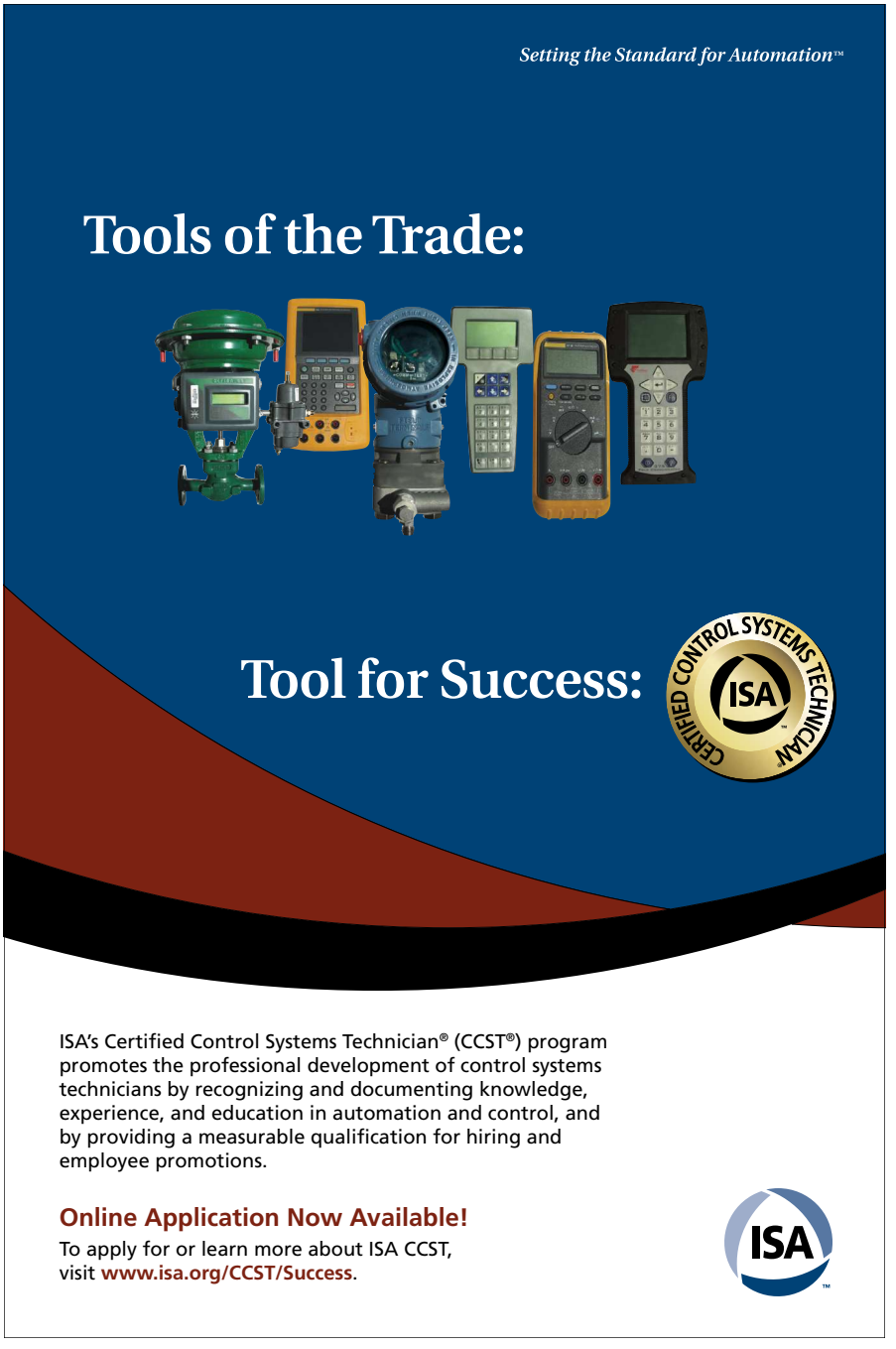
The CAP body of knowledge covers a vast set of knowledge areas, including control systems design, installation, maintenance, and so on.

With automation becoming ubiquitous in industries, the automation professional encounters a wide range of automated control systems from batch processing in food industries to continuous critical and safety processes in the oil and gas sectors, and it becomes incumbent upon the automation professional to keep themselves familiarized with industry requirements.

To meet these growing demands, the CAP body of knowledge (www.isa.org/link/CAP_skills) covers a wide range of topics from MES system design, enterprise control systems, safety instrumented systems and batch processing.

With the integration of automation as

an integral part in our production industries and growing demands, standardization and need for skilled recourses has now become a proven fact. CAP helped me to match up with these demands. It increased my reach to new job markets and advocated as a potential resource looking ahead to grow in the automation field. Peers tend to find CAP unique and grow curious, so it provided me a great opportunity to increase my professional networking across the field.



ABOUT THE AUTHOR

Abdul Rauf (arauf@avanceon.com) is an ISA Certified Automation Professional working at Avanceon as senior application engineer.

REFERENCES

Certified Automation Professional (CAP)
www.isa.org/cap

A Guide to the Automation Body of Knowledge, Second Edition
www.isa.org/autobok

In memoriam

James Patrick Carew of Cinnaminson, N.J., passed away 11 August 2011. He was a native of Brooklyn, N.Y. and formerly lived in New York, New Jersey, California, and Texas. Carew served in the U.S. Army, stationed in Georgia and Fort Dix during the Korean War.

Carew was a Life Senior Member of ISA, joining in 1969. He was a member of the Houston Section for many years. He most recently worked on several ISA standards, including on ISA5, Documentation of Measurement and Control Instruments and Systems (voting member); ISA5.1, Instrumentation Symbols and Identification (voting member and chair); ISA5.8, Measurement & Control Terminology Review (voting member); ISA84, Electrical/Electronic/Programmable Electronic Systems (E/E/PES) for Use in Process Safety Applications (information member; previously voting member); and ISA97, In-Line Sensors (voting member).

Carew was a retired engineer for Stone & Webster, Boston, Mass., where he designed chemical plants and refineries. He was a graduate of The Polytechnic Institute of Brooklyn with a Bachelor of Science in Mechanical Engineering degree.

Documenting skills is value-add

ISA certification provides an objective, third-party assessment, and confirmation of a person's skills. It gives manufacturing and factory staff the opportunity to differentiate themselves from their peers and gain recognition. *InTech* covers two certification areas in this monthly Certification department.

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

CAP question

Which of the following network security technologies does not use encryption?

A. Digital Signatures
B. Virtual Local Area Network (VLAN)
C. Virtual Private Network (VPN)
A. Wired Equivalent Privacy (WEP)

CAP answer

The correct answer is **B**. A virtual local area network (VLAN) is achieved through configuration of Ethernet switches and not encryption. It divides a physical network into smaller logical networks to increase performance, improve manageability, and simplify network design.

Answer A is not correct because digital signatures are encrypted. Information signed, such as a contract or data record, is compressed via a cryptographic one-way function (a hash) into a short bit string.

Answer C is not correct because a virtual private network (VPN) is an encrypted private network that operates as an overlay on a public infrastructure.

Answer D is not correct because Wired Equivalent Privacy (WEP) is an encryption algorithm for IEEE 802.11 wireless networks, originally intended to provide data confidentiality comparable to a wired network.

Reference: ANSI/ISA-TR99.00.01-2400 Security Technologies for Manufacturing and Control Systems

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CCST question

A fluid is flowing through a 10-inch diameter pipe at a velocity of 6 feet/sec. When the pipe reduces to an 8-inch diameter, and all other flowing parameters remain the same, the fluid velocity becomes _____ feet/sec.

A. 2.550
B. 6.075
C. 9.375
D. 12.75

CCST answer

The "constant" between the run of pipe that has a 10-inch diameter and the run of pipe with a diameter of 8 inches is that the flow rate (and other fluid properties) is the same in both pipe lengths.

Flow through a round pipe can be expressed as:

$Q = \text{Velocity (ft/sec)} \times \text{Area of Pipe (ft}^2\text{)}$

Since flow is constant between the two pipe sizes, we can set:

$\text{Velocity}_1 \times \text{Area}_1 = \text{Velocity}_2 \times \text{Area}_2$,

where the subscript = 1 for the initial conditions (10-inch pipe) and subscript = 2 for the final conditions (8-inch pipe).

Solving for Velocity_2:

$\text{Velocity}_2 = \text{Velocity}_1 \times \text{Area}_1 / \text{Area}_2$

For the 10-inch pipe, Area (in ft^2) = pi x $D^2 / 4 = 0.5454\ ft^2$. Don't forget to divide 10 inches by 12 to get Diameter in feet before "squaring."

For the 8-inch pipe, Area (in ft^2) = pi x $D^2 / 4 = 0.3490\ ft^2$

Substituting these values in the equation for Velocity_2 above:

$\text{Velocity}_2 = \text{Velocity}_1 \times \text{Area}_1 / \text{Area}_2$

$\text{Velocity}_2 = 6.0 \text{ ft/sec} \times 0.5454 / 0.3490$
$= 9.375 \text{ ft/sec}$

The correct answer is **C**.

Reference: Goettsche, L.D. (Editor), *Maintenance of Instruments and Systems, 2nd Edition* (2005), ISA Press

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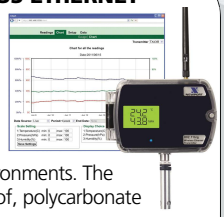

NEWPORT® unveils new wSeries Wi-Fi transmitters. The wSeries transmitters communicate on a standard Wi-Fi network which is an ideal and economical solution for facilities with an existing Wi-Fi network as well as new installations. The wSeries Wi-Fi sensors are designed for demanding industrial applications and harsh outdoor environments. The electronics are protected in a rugged weatherproof, polycarbonate NEMA4/IP65 rated housing.

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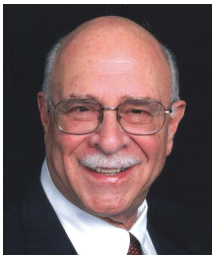
Why is good control important?

By Dick Caro



It takes a lot of effort to attain "good control." Certainly, loop tuning is taught at ISA, but few control/instrument engineers actually practice loop tuning. Most of the distributed control systems (DCSs) have some type of automated loop tuning, and suppliers provide loop tuning software that can be added to any system, including programmable logic controllers (PLCs). Some processes, however, make it almost impossible to achieve optimal loop tuning. Understanding ways to improve control makes engineers more valuable to their company and improves operations.

First, what do we mean by "good control?" A control loop is said to be "in tune" when the process does not deviate from the setpoint. A poorly tuned control loop has excessive deviations from setpoint, which is referred to as noise, often caused by improperly set tuning constants. Sometimes, the process is nonlinear, so one set of optimal tuning constants to have good control is not possible. Processes that have large dead times such that the process reaction time to setpoint changes or load changes are greater than the process time constant cannot be tuned to improve performance using simple feedback control are another challenge. In these cases, improvements can be achieved by selecting some type of advanced control.

Aside from simple feedback loop control concepts, "good control" also means the process setpoints are set at the values that will make the process economic. Every process has some economics that should govern the values for all of the setpoints. The objectives of running the process will determine those setpoints. For example, the setpoints for achieving maximum product production rate are quite different from those to achieve minimal consumption of energy. Often, the product purity is the governing factor, while in other processes, it is more economical to assure that the least amount of a product component winds up in the waste stream or in a secondary product flow. Many times, the primary economic factor is product yield.

In many processes, the operating conditions must be held within certain constraints. Very often, the process may be operated anywhere between those constraints, but some type of penalty will result if those constraints are violated. Processes with a high amount of variability must have their setpoints adjusted such that the constraint will not be violated.

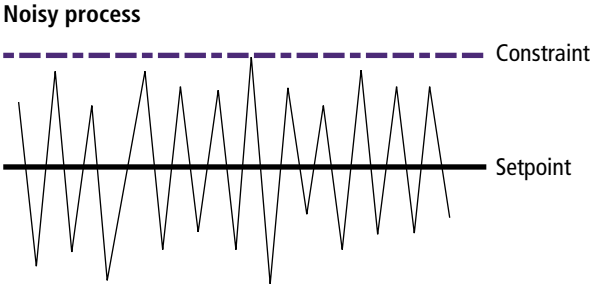
Control loops that are out of tune or do not have some type of advanced control to eliminate or compensate for long dead time, appear to be "noisy," as illustrated by the left-hand recording chart in the figure. When optimal tuning or advanced control is implemented, the recording chart might appear more like that illustrated by the right-hand chart in the figure.

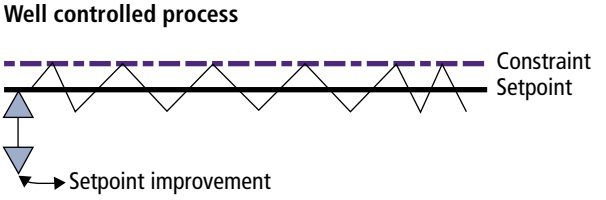
Notice the process constraint remains the same, but the reduced process variance now allows us to raise the setpoint much closer to that constraint. This assumes that by raising the setpoint closer to the constraint, some economic advantage will be attained, which here is called "setpoint improvement." The setpoint improvement might be to increase production rate or to improve product yield, for example.

Note it is not enough just to tune the control loop to gain this economic advantage. You must adjust the setpoints. Of course, it was the reduction in control loop variability that enabled you to adjust the setpoint. Now you know why good control is important.

ABOUT THE AUTHOR

Dick Caro is an industrial automation consultant for CMC Associates, a Certified Automation Professional, and a Life Fellow of ISA. Caro chairs the ISA50 (fieldbus) standards committee and two ISA100 (wireless) subcommittees.





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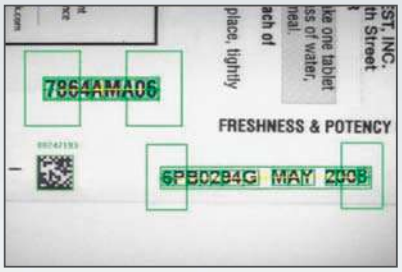

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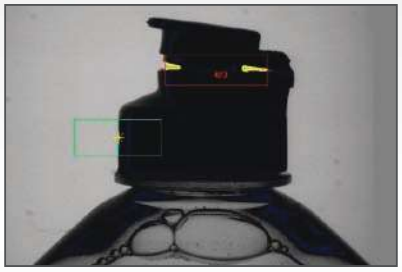
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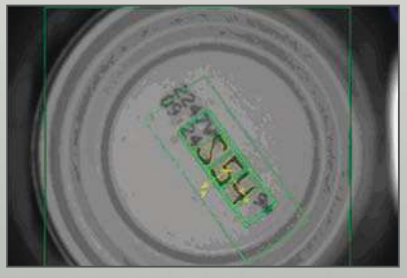
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